REVIEW

Video anomaly detection in 10 years: a survey and outlook

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Abstract
Video anomaly detection (VAD) holds immense importance across diverse domains such as surveillance, healthcare, and environmental monitoring. While numerous surveys focus on conventional VAD methods, they often lack depth in exploring specific approaches and emerging trends. This survey explores deep learning-based VAD, expanding beyond traditional supervised training paradigms to encompass emerging weakly supervised, self-supervised, semi-supervised, and unsupervised approaches. A prominent feature of this review is the investigation of core challenges within the VAD paradigms, including large-scale datasets, feature extraction, learning methods, loss functions, regularization, and anomaly score prediction. Moreover, this review investigates vision-language models (VLMs) as potent feature extractors for VAD. VLMs integrate visual data with textual descriptions from videos, enabling a nuanced understanding of scenes crucial for anomaly detection. By addressing these challenges and proposing future research directions, this review aims to foster the development of robust and efficient VAD systems leveraging the capabilities of VLMs for enhanced anomaly detection in complex real-world scenarios. This comprehensive analysis seeks to bridge existing knowledge gaps, provide researchers with valuable insights, and contribute to shaping the future of VAD research.

Keywords Video anomaly detection · Vision-language models · Deep learning · Weakly supervised learning

1 Introduction

Anomaly detection refers to the task of identifying events or patterns that deviate from expected behavior within a specific data context [1, 2]. It plays a central role in several applications such as fraud detection, cybersecurity for network intrusion detection, quality assurance in manufacturing, fault identification in industrial machinery [3], healthcare monitoring [4], environmental monitoring [5], autonomous vehicle control [6], and beyond. However, our primary focus is on video anomaly detection (VAD) [7], which is a key technology that automates the identification of unusual or suspicious activities within video sequences.

Although conventional methods for VAD have been extensively explored, the rise of deep learning has opened up new avenues in the VAD field [8]. Deep learning algorithms, such as convolutional neural networks (CNNs) and vision transformers (ViTs), have shown impressive capabilities in extracting complex patterns and representations from large-scale data [9, 10], leading to more accurate and reliable detection of anomalies in video data. These advances have addressed many limitations of conventional machine learning methods, including their reliance on handcrafted features [11, 12]. Most importantly, not all recent methods rely on fully labeled datasets (supervised learning). There has been growing interest in weakly supervised, semi-supervised, self-supervised, and unsupervised learning paradigms [13–15]. These alternative approaches present promising solutions to the

challenges encountered by conventional VAD methods, such as the requirement for extensively annotated datasets and the complexity of capturing spatiotemporal patterns. Using such strategies, researchers are designing more robust VAD models that can adapt across various scenarios [1, 16].

Similarly, several research challenges remain, one of the most pressing is the *quality and diversity of available datasets*. Therefore, the development of strong and comprehensive datasets such as the UCF-Crime dataset [15], XD-Violence dataset [17], and ShanghaiTech dataset [18] has played a pivotal role in advancing the field by capturing a wide range of real-world anomalies. In VAD, videos consist of sequences of frames, resulting in complex, high-dimensional spatiotemporal data. Detecting anomalies within such complex data necessitates methods capable of *effectively capturing spatial, temporal, spatiotemporal, and textual features*. To address these challenges, numerous VAD methods and deep feature extractors have been introduced in academic research, which has played a significant role in advancing the current state-of-the-art. It is also worth mentioning the importance of choosing a correct loss function based on the type of VAD paradigm. Hence, *loss functions and regularization* techniques are another challenge in the VAD problem. Another significant challenge lies in the diverse paradigms, including *self-supervised, semi-supervised, weakly supervised, fully supervised, and unsupervised models* utilized to address the VAD problem.

These paradigms can be categorized into classical and deep learning. The classical approaches employ handcrafted features, including spatiotemporal gradient [19], histograms of oriented gradients (HOG) features [20, 21], and histograms of optical flows (HOF) [22] within a temporal cuboid. These features are selected for their effectiveness in capturing appearance and motion information in a spatiotemporal context.

The deep learning approaches are more powerful in terms of rich feature representation and end-to-end learning and have gained much popularity in the past decade due to advancements in different learning models. Specifically, these approaches leverage powerful feature extractors to capture meaningful spatiotemporal features. Examples include convolutional neural networks [23], autoencoders [24], GANs [25], vision transformers [26, 27], and vision-language models [28, 29]. In this survey, we also identify these different models and provide take-home messages for each of the aforementioned challenges.

To illustrate the impact of deep learning advancements in VAD, Fig. 1 highlights the performance variation of representative deep learning-based VAD methods on two widely recognized benchmark datasets: UCF-Crime [15] and ShanghaiTech [18].

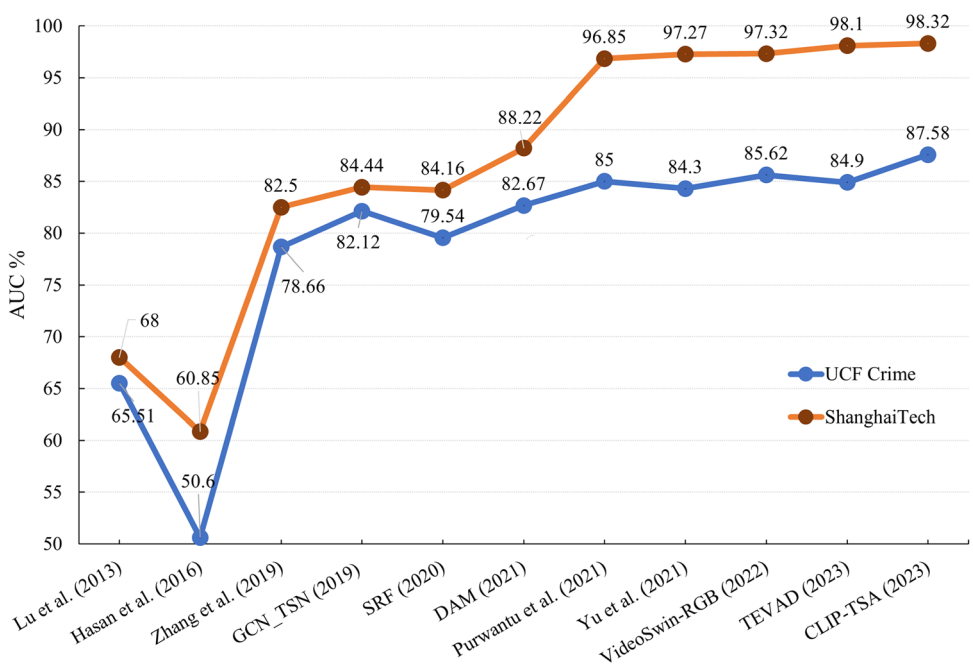

Fig. 1 Performance improvement from 2017 until 2023 on two popular benchmarks. Performance is measured by the area under the ROC curve (AUC%). Note that some of the models were developed before the datasets were created but were used after the creation by other researchers such as [19, 20], the other proposed models are: [27, 30–37]

The performance is reported in terms of the area under the ROC curve (AUC%). It can be observed from the figure that substantially high values of the AUC score were achieved by recent deep learning-based models in the literature, which include a score of 87.58% by the CLIP-TSA [30] model on the UCF-Crime dataset and a score of 98.10% by the TEVAD [31] model on the ShanghaiTech dataset. The performance improvement trend in this figure underscores a significant advancement in deep learning methodologies over the past decade, with the most notable improvement achieved by utilizing the recently proposed vision-language-based model [30].

Motivation: Despite the growing interest in VAD and the proliferation of deep learning-based approaches, there remains a need for a comprehensive survey that explores the latest developments in this field. Existing surveys often focus on traditional VAD methods and may not adequately cover the emerging trends and methodologies such as vision-language models (VLMs) [30, 31, 38–40]. By delving into VLMs and deep learning-based VAD, including supervised, semi-supervised, weakly supervised, self-supervised, and unsupervised approaches, this survey aims to provide a thorough understanding of the state-of-the-art techniques and their potential applications. Our main focus lies on deep learning-based solutions for VAD problems. Particularly, we explore emerging VAD paradigms that employ different datasets, feature extractions, loss functions, and spatiotemporal regularization. In addition, we offer a comprehensive analysis of over 80 different methodologies employed in VAD, with a specific focus on the potential of textual features extracted with vision-language models. In this survey, our main contributions are summarized as follows:

- We identify the core challenges of the SOTA VAD paradigms, including the vision-language models in terms of feature extraction, large-scale VAD training datasets, loss functions, spatiotemporal regularization, and video anomaly scores prediction.
- We provide a comparative analysis through quantitative and qualitative comparisons of SOTA models on different benchmarking datasets. This analysis sheds light on the strengths and weaknesses of existing methodologies, offering valuable insights for researchers and practitioners in the field.
- Following our analysis, we outline our proposed recommendations for addressing the open challenges in VAD. These recommendations are informed by our deep understanding of the current landscape of VAD research and aim to guide future research directions toward overcoming existing limitations and advancing the SOTA.

The structure of this paper is as follows: Sect. 2 outlines the methodology used to select the research studies included in this review over the past decade. Section 3 examines previous surveys conducted in the field of video anomaly detection. Section 4 presents the formulation of the video anomaly detection problem. Section 5 presents a systematic approach and taxonomy for analyzing the VAD problem, and then the core challenges including the datasets Sect. 5.1, feature extraction Sect. 5.2, supervision schemes Sect. 5.3, loss functions Sect. 5.4, regularization techniques Sect. 5.5, and the anomaly score Sect. 5.6. Section 6 outlines the datasets' guidelines and evaluation protocols used. Section 7 provides a comparative analysis through quantitative and qualitative comparisons of state-of-the-art models. Section 8 provides a visualization of bibliometric networks for thematic analysis. Finally, we conclude our work and present additional future directions in Sect. 9.

2 Review methodology

This survey paper exclusively considers research directly related to "video anomaly detection." The survey is conducted systematically, utilizing publications from top computer vision venues, including CVPR (Conference on Computer Vision and Pattern Recognition), ICCV (International Conference on Computer Vision), ECCV (European Conference on Computer Vision), IEEE TPAMI (Transactions on Pattern Analysis and Machine Intelligence), IJCV (International Journal of Computer Vision), CVIU (Computer Vision and Image Understanding), AAAI Conference on Artificial Intelligence, EAAI (Engineering Applications of Artificial Intelligence), IEEE TII (IEEE Transactions on Industrial Informatics), IEEE TCSVT (IEEE Transactions on Circuits

and Systems for Video Technology), ICIP (International Conference on Image Processing), IEEE AVSS (Conference on Advanced Video and Signal Based Surveillance), NCAA (Neural Computing and Applications), and IVC (Image and Vision Computing). The objective of this review paper is to present the state-of-the-art in video anomaly detection. Accordingly, this study focuses on identifying published research concerning the implementation of various computer vision-based methods and deep learning to address the challenges of VAD. The reviewed works span the last decade, covering around 80 articles.

3 Related work

In the field of anomaly detection, several survey papers have been conducted in the past decade [1, 8, 11, 12, 41–44]. The first survey paper, published in 2018 by Kiran et al. [12], focused on deep learning techniques for VAD, with particular attention to unsupervised and semi-supervised methods. The authors contributed to the understanding of early VAD architectures by classifying models into three main categories: reconstruction-based, spatiotemporal predictive, and generative models. They also presented a technical analysis of how key models operate, including conventional autoencoders (CAEs), LSTMs, GANs, and PCA. Moreover, their benchmarking across standard datasets such as UCSD and CUHK Avenue facilitated direct performance comparisons. However, certain limitations become apparent when viewed from today's perspective. Since the paper was published before the rise of multiple instance learning (MIL), it did not cover the emerging weakly supervised approach. It also did not address more recent advances, such as vision-language models and transformer architectures, which have significantly shaped the current VAD field.

Another significant study by Chalapathy et al. [41] highlighted the potential of deep anomaly detection methods to address various detection challenges. Their work covered diverse application areas such as the Internet of Things, intrusion detection, and surveillance videos. One of the paper's strengths is its clear categorization of both anomaly types, collective, contextual, and point anomalies, and deep learning approaches, including unsupervised, semi-supervised, hybrid, and one-class neural network models. This structure helps readers grasp the broader understanding of anomaly detection. The inclusion of practical applications enhances the relevance of the work, especially for those aiming to apply these techniques in real-world domains. However, since their focus was on general anomaly detection, the paper did not go through the specific challenges of video anomaly detection. Hence, it did not provide strong discussion on key aspects like modeling temporal dependencies, capturing motion patterns, and handling anomalies over time. In addition, standard video datasets and evaluation metrics commonly used in VAD research, such as frame-level AUC, are also not covered. Moreover, as the paper was published before the new approaches like transformers, vision-language models, and MIL-based weak supervision, it does not reflect many of the recent trends shaping the VAD field today.

In another survey, Ramachandra et al. [1] focus primarily on detecting anomalies within a single scene while also highlighting the differences from multi-scene anomaly detection. An important distinction lies in the fact that single-scene VAD may involve anomalies dependent on specific locations, whereas multi-scene detection cannot. The survey also sheds light on benchmark datasets employed for single-scene versus multi-scene detection and the associated evaluation procedures. In a broader context, the survey classifies previous research in video anomaly detection into three main categories: distance-based, probabilistic, and reconstruction-based approaches. This taxonomy, along with the comparative analysis of different methods across accuracy and equal error rate, provides a structured foundation of the VAD field. Additionally, the survey offers a critical assessment of evaluation metrics, where they have also introduced region- and track-based ROC curves to address the shortcomings of traditional frame and pixel-level metrics. Finally, it concludes with a set of best practices, including the use of multiple datasets, reporting inference speed, and visualizing false positives for reproducibility. However, the survey predates the rise of transformer-based and multi-modality models that have since reshaped the field. Its discussion of multi-scene anomaly detection is brief and conceptual, without methodological depth.

Additionally, although they have provided a comprehensive overview of the existing datasets, their overview remains descriptive and does not delve into annotation biases or fairness considerations.

In a different work, Nayak et al. [11] categorize learning frameworks into four main categories: supervised, unsupervised, semi-supervised, and active learning. They have also classified the deep learning-based VAD methods into several distinct categories, including trajectory-based methods, global pattern-based methods, grid pattern-based methods, representation learning models, discriminative models, predictive models, deep generative models, deep one-class deep neural networks, and deep hybrid models. Moreover, the research provides a comprehensive analysis of performance evaluation methodologies, covering aspects such as the choice of datasets, computational infrastructure, evaluation criteria, and performance metrics. Building on these contributions, the survey has several strengths, where it provides a dual-perspective analysis of the deployment goals: explicitly contrasting accuracy-oriented (offline applications) and processing-time-oriented (real-time applications) approaches; discussing where each excels or breaks down. Importantly, the survey concludes with a comprehensive discussion of key research challenges, including the need for better datasets, annotation scarcity, high computational complexity, evaluation methodology gaps, and the difficulty of achieving accurate yet real-time detection. However, the survey predates recent advances such as multi-modalities for VAD, transformer-based architectures, vision-language models, and MIL-based weak supervision, limiting its coverage of the latest methodological trends in VAD. Additionally, it lacks a detailed analysis of loss functions, providing only scattered references tied to individual methods without offering comparative analysis or design rationale.

Another work by Pang et al. [8] focused on deep learning techniques for anomaly detection and explored various challenges within the anomaly detection (AD) problem. These challenges included issues such as class imbalance in the data, complex anomaly detection, the presence of noisy instances in weakly supervised AD methods, and more. They discussed how deep learning methods offer solutions to these diverse challenges. To structure their analysis, Pang et al. introduced a hierarchical taxonomy for categorizing deep anomaly detection methods. This taxonomy comprised three principal categories: deep learning for feature extraction, learning feature representations of normality, and end-to-end anomaly score learning. Additionally, they provided 11 fine-grained subcategories from a modeling perspective. In addition to that, the survey offers several strengths. The authors enrich their analysis with practical resources, including summaries of over 30 representative algorithms, 21 publicly available datasets, and links to open-source implementations for anomaly detection in general. They also outline key future research directions, including weak supervision, few-shot anomaly detection, and learning transferable pretrained representation models from large-scale unlabeled data in an unsupervised or self-supervised manner. However, the survey predates recent advances such as transformer-based architectures and vision-language models. Its coverage of video-specific challenges, such as temporal modeling and anomaly localization, is minimal, and it lacks unified quantitative performance comparisons.

In their research, Mohammad Baradaran and Robert Bergevin [42] delve deeply into the semi-supervised VAD approaches, focusing on scenarios where labeled anomaly data are limited. They emphasize the role of feature extractors where they offer a comparative view of how different architectures, such as CNNs, 3D CNNs, ConvLSTMs, GANs, and memory modules, have a strong ability to capture crucial spatial and temporal representations in semi-supervised settings.

Another key strength of the paper lies in proposing a clear taxonomy comprising six methodological categories: reconstruction, prediction, memorization, object-centric, segmentation, and multitask learning. Additionally, they have provided an experimental analysis of existing methods, where they have examined the instances where these methods fail to illuminate the respective strengths and weaknesses of different approaches. However, the paper has limitations. It predates more recent developments such as transformer-based architectures and vision-language models for semi-supervised learning. Additionally, while the paper discusses the performance of various methods qualitatively, it does not present any numerical comparisons, such as AUC scores across multiple methods using standard datasets. There is no unified benchmark table or quantitative analysis that allows for direct performance comparison of the surveyed approaches. Moreover, the paper offers only limited

discussion on the design of loss functions, fairness of datasets issues, and practical deployment constraints like computational efficiency or suitability for real-time systems.

Another recently published survey paper by Nomica et al. [43] provides an analysis of machine learning techniques for detecting anomalies in video surveillance systems. It categorizes these methods into supervised, semi-supervised, and unsupervised approaches, highlighting their strengths, weaknesses, and applicability. The survey includes comparisons that summarize representative algorithms, public datasets, and commonly used evaluation metrics, and it outlines key research challenges such as data scarcity, privacy concerns, and real-time performance. However, the survey remains largely descriptive and does not offer a unified quantitative benchmark for directly comparing methods across standard datasets using consistent evaluation metrics. Moreover, it does not explicitly differentiate between feature modalities, such as spatial, temporal, textual, or hybrid features, nor does it analyze how the choice of feature extractor influences anomaly detection performance. The review also omits detailed coverage of transformer-based architectures and entirely overlooks the emerging class of vision-language models, which are increasingly influential in multimodal anomaly detection. Additionally, the discussion on datasets is limited and lacks a clear comparative analysis of their characteristics, challenges, or suitability for different VAD paradigms.

A very recently published survey paper in 2024 by Yang et al. [44] categorizes the VAD approaches into unsupervised, weakly supervised, fully unsupervised, and supervised VAD. It highlights their strengths, such as improved feature extraction and detailed object analysis, while also noting the importance of handling spatiotemporal features and illumination changes. Among its key strengths, the paper introduces a unified intuitive generalized video anomaly event detection (GVAED) taxonomy based on the different supervision schemes, model input data and network structure, it provides a quantitative performance benchmark across multiple datasets, and offers a well-maintained GitHub repository with links to papers, evaluation metrics, and datasets, making it highly accessible for new researchers. It also outlines a clear set of open research challenges, including generalization, domain adaptation, and lightweight deployment. However, despite the growing trend of the VLM as feature extractors, they did not emphasize the huge importance of the VLM for multimodal models. The issue of dataset bias, annotation noise, and fairness is acknowledged only briefly.

To conclude, while previous survey papers have provided valuable foundations for understanding various aspects of video anomaly detection, they have several recurring limitations. In particular, earlier surveys often overlook the importance of feature modality differentiation. The role of textual features and their integration with vision-language models has also been neglected, despite their growing importance in multimodal video anomaly detection. Additionally, many prior works lack quantitative benchmarks across the standard datasets, offering only qualitative evaluations, which limit direct performance comparisons. Other gaps include minimal coverage of transformer-based architectures, or analysis of loss function design, limited consideration of dataset bias, fairness, and real-time deployment constraints, alongside the absence of unified taxonomies that comprehensively connect learning paradigms with feature extraction strategies.

Our survey addresses these gaps by emphasizing the effect of using VLMs as powerful feature extractors, which are capable of bridging the semantic gap between visual and textual modalities. Unlike previous works, we provide a clear and comprehensive taxonomy that categorizes VAD methods based on both learning/supervision strategies (supervised, unsupervised, self-supervised, and weakly supervised) and feature extraction approaches. We also offer practical insights into loss function and regularization techniques, along with dataset selection guidelines and evaluation metrics tailored to the VAD domain. Moreover, we include both quantitative and qualitative comparative analysis of state-of-the-art methods, giving the readers a more complete picture of current performance trends. Finally, an overview comparison between our survey paper and previous surveys is summarized in Table 1, which highlights our main contributions in the VAD domain.

3.1 Contributions

In this survey, our main contributions compared to existing survey papers are summarized as follows:

Table 1 Comparison of previous surveys on video anomaly detection (VAD)

Survey	Sup	Semi	Weak	Self	VLMs	Deep	Eval	Loss	Comp
Kiran et al. (2018) [12]	✗	✔	✗	✗	✗	✔	✔	✔	✔
Chalapathy et al. (2019) [41]	✔	✔	✗	✗	✗	✔	✗	✗	✗
Ramachandra et al. (2020) [1]	✔	✔	✔	✗	✗	✔	✔	✗	✔
Nayak et al. (2021) [11]	✔	✔	✗	✗	✗	✔	✔	✔	✔
Pang et al. (2021) [8]	✗	✔	✔	✔	✗	✔	✗	✔	✔
Baradaran & Bergevin(2024) [42]	✗	✔	✗	✗	✗	✔	✗	✗	✔
Nomica et al. (2024) [43]	✔	✔	✔	✗	✗	✔	✗	✗	✗
Yang et al. (2024) [44]	✔	✔	✔	✗	✗	✔	✔	✔	✔
Our survey (2024)	✔	✔	✔	✔	✔	✔	✔	✔	✔

Sup.—Supervised, Semi—Semi-Supervised, Weak—Weakly Supervised, Self—Self-Supervised, VLMs—Vision-Language Models, Deep—Deep Feature Extraction, Eval.—Evaluation Metrics, Loss—Loss Functions, Comp.—Comparative Analysis

- We define a clear and comprehensive problem formulation of the VAD problem, specifically in the context of supervised learning, where frame-level labels are available.
- To the best of our knowledge, this is the first survey to highlight the emerging importance of integrating vision-language models (VLMs) as feature extraction in VAD. We explore how VLMs can significantly enhance model performance by effectively combining visual and textual data to better understand and detect anomalies.
- The paper is organized to serve as a detailed roadmap for readers and researchers new to the field of VAD. We provide foundational knowledge and a structured framework to navigate the complexities of VAD research as shown in Fig. 2.
- We present a well-structured discussion starting from the selection of datasets and the types of features that are critical for VAD. This includes a focus on textual features and deep feature extractors. We also explore the various learning and supervision paradigms, including supervised, self-supervised, weakly supervised, unsupervised, and semi-supervised (reconstruction and prediction) approaches, detailing their advantages and disadvantages.
- We provide a systematic taxonomy of video anomaly detection, which is categorized into two main dimensions: learning and supervision schemes, and feature extraction as shown in Fig. 3.

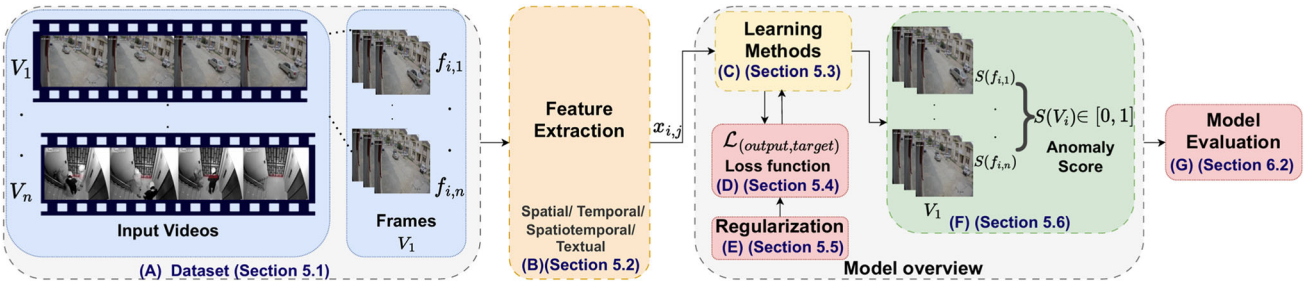

Fig. 2 End-to-end **video anomaly detection** pipeline that also serves as the organizing backbone of this survey, covering the process from dataset construction to model evaluation. **A Dataset (Sect.** 5.1): a set of surveillance videos is split into individual frames. **B Feature Extraction (Sect.** 5.2): Spatial, temporal, spatiotemporal, or textual features are computed using feature extractors. **C Learning Methods (Sect.** 5.3): The extracted features are fed into one of the supervision schemes. **D Loss Functions (Sect.** 5.4) and **E Regularization Integration (Sect.** 5.5): Guide training and improve model generalization. **F Anomaly-Score Computation (Sect.** 5.6): The trained model outputs a continuous anomaly score for each frame or clip. **G Model Evaluation (Sect.** 6.2): These scores are assessed using standard evaluation metrics. The colored blocks map one-to-one to the survey subsections listed in parentheses

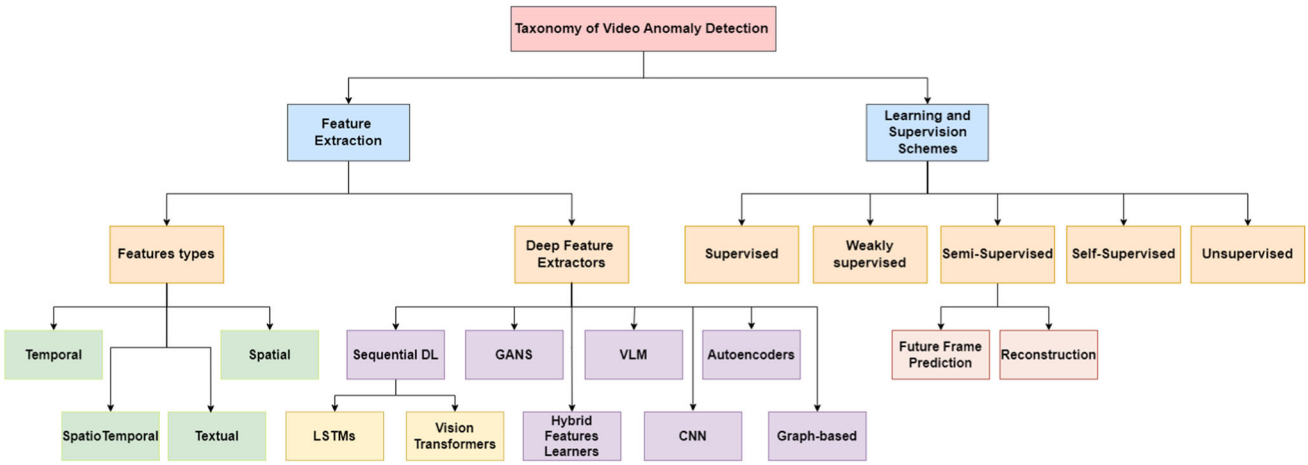

Fig. 3 Taxonomy of video anomaly detection

- We provide insights about the selection of the appropriate loss functions and regularization techniques, which are crucial for optimizing the performance of VAD models.
- We introduce a guideline to help researchers select the most appropriate datasets and evaluation metrics for their experiments, making it easier to fairly assess and compare different VAD methods.
- We offer a detailed comparative analysis of the state-of-the-art (SOTA) VAD methods in both quantitative and qualitative terms to help in understanding the current landscape and performance benchmarks in the field.
- Finally, in this work, we provide an extensive discussion on potential future research directions in the VAD field. This includes exploring new technologies, methodologies, and application areas, and guiding researchers toward the investigation of future avenues.

4 Defining the video anomaly detection problem

The primary goal of video anomaly detection is to implement resilient algorithms and models capable of automatically identifying and flagging anomalies in video data that deviate from the regular pattern. This involves converting raw video data into interpretable feature representations by utilizing robust feature extractors capable of capturing both spatial and temporal characteristics. Additionally, it involves the selection of appropriate algorithms or techniques and the establishment of effective evaluation metrics to accurately assess the detection performance.

In the **supervised** learning setup, during training and testing, both normal and anomalous videos are included, using frame-level labels (Y_{v_i}) to distinguish between normal and anomalous frames. In this context, where frame-level labels are available, the video anomaly detection problem can be described as follows:

Each video is represented as V_i, which consists of a sequence of consecutive frames $\{f_{i,1}, f_{i,2}, \ldots, f_{i,n}\}$. From each frame, we can extract essential feature representations, denoted by $x_{i,j}$. Define a model M that takes these features extracted from each frame and produces an anomaly score for that frame. For each frame $f_{i,j}$, the anomaly score is:

$$S(f_{i,j}) = M(x_{i,j}) \tag{1}$$

This anomaly score $S(f_{i,j})$ is compared with a predetermined threshold T; we can define the predicted binary label, $\hat{Y}_{f_{i,j}}$, for each frame as:

$$\hat{Y}_{f_{i,j}} = \begin{cases} 1 & \text{if } S(f_{i,j}) \geq T \\ 0 & \text{if } S(f_{i,j}) < T \end{cases}$$

Where: 1 indicates that $f_{i,j}$ is anomalous and 0 indicates that $f_{i,j}$ is normal. The true label of each frame is represented as $Y_{f_{i,j}} \in \{0, 1\}$. The objective is to train the model M such that the difference between $\hat{Y_{v_i}}$ and Y_{v_i} is minimized for all video frames in the training set and generalizes well to unseen videos.

5 Analyzing video anomaly detection: a systematic approach

In this section, we dig deeper into the complexities of the VAD process by analyzing relevant literature. Figure 2 serves as a visual roadmap for readers and researchers new to the field of VAD. It also structures the remainder of the survey by illustrating how the VAD pipeline operates (i.e., through a left-to-right data flow), starting from raw surveillance footage and progressing through to the final computation of anomaly scores. We begin with a collection of surveillance videos (block A; see Sect. 5.1). Each video is split into individual frames, which are then passed to the feature extraction process (block B; see Sect. 5.2), where spatial, temporal, spatiotemporal, or textual features are extracted, leveraging deep CNNs, sequential models, and generative models, or modern vision-language models. The resulting features pass into the learning module (block C; see Sect. 5.3), where an appropriate supervision scheme (supervised, weakly supervised, semi-supervised, self-supervised, or unsupervised) is applied. Training is guided by the chosen loss function for that scheme (block D; see Sect. 5.4) and by regularization techniques that improve generalization (block E; see Sect. 5.5). The trained model produces a continuous anomaly score (block F; see Sect. 5.6) for each frame or clip, indicating the likelihood of abnormality, and these scores are finally evaluated with the protocols summarized in Sect. 6.2 (block G). This pipeline, therefore, acts as a visual roadmap; each colored block corresponds to a dedicated subsection in Sect. 5, which guides the reader from raw video data to quantitative evaluation.

5.1 Taxonomy of video anomaly detection

The taxonomy of video anomaly detection, as shown in Fig. 3, is systematically categorized into two main dimensions: learning and supervision schemes (see Sect. 5.3) and feature extraction (see Sect. 5.2). The learning and supervision schemes dimension includes supervised, self-supervised, weakly supervised (such as multiple instance learning), unsupervised methods, and semi-supervised, including reconstruction and prediction approaches. The feature extraction dimension involves deep feature extractors, including CNNs, autoencoders, GANs, sequential deep learning models (like LSTMs and vision transformers), vision-language models, graph-based models, and hybrid models. Additionally, it covers different types of features such as spatial, temporal, spatiotemporal, and textual.

5.2 Datasets building and selection

The field of video anomaly detection relies significantly on publicly available datasets, which are used for testing and benchmarking the proposed models. In this section, we present an overview of the common datasets utilized in the field of VAD, each carefully curated to facilitate the study of anomalies in various scenarios. These datasets encompass a wide range of scenes, offering diverse challenges for anomaly detection. We will investigate the number of videos contained in each dataset, the specific types of scenes they cover, and the anomalies present.

5.2.1 Subway dataset

The subway dataset [45] consists of two videos that were collected using a CCTV camera, each capturing distinct perspectives of an underground train station. The first video of the dataset focuses on the "entrance gate" area, where individuals are typically descend through the turnstiles and entering the platform with their backs turned to the camera. On the other hand, the second video is at the "exit gate," observing the platform where passengers ascend while facing the camera. The dataset has a combined duration of 2 h. Anomalies within this dataset encompass activities such as the movement of pedestrians in the wrong direction. Notably, this dataset is recorded within an indoor environment.

5.2.2 UCSD pedestrian

The UCSD anomaly detection dataset was proposed by Mahadevan et al. [46] and collected using a stationary camera placed at an elevated vantage point to monitor the pedestrians' walkways. This dataset offers a wide range of scenes depicting varying crowd densities, spanning from sparsely populated to highly congested environments. Normal videos within the dataset predominantly feature pedestrians, while abnormal events arise from two primary sources: the presence of non-pedestrian entities in the walkways and the occurrence of anomalous pedestrian motion patterns. Common anomalies observed include bikers, skaters, small carts, individuals walking across designated walkways or in adjacent grass areas, as well as instances of people in wheelchairs. The dataset comprises two distinct subsets, Peds1 and Peds2, each capturing different scenes. Peds1 depicts groups of people walking toward and away from the camera, often exhibiting perspective distortion, while Peds2 focuses on scenes where pedestrians move parallel to the camera plane.

The dataset is further divided into clips, with each clip containing approximately 200 frames. Ground truth annotations are provided at a frame level, indicating the presence or absence of anomalies. Additionally, a subset of clips in both Peds1 and Peds2 is accompanied by manually generated pixel-level binary masks, enabling the evaluation of algorithms' ability to localize anomalies. The UCSD Ped1 & Ped2 datasets are publicly available: http://www.svcl.ucsd.edu/projects/anomaly/.

5.2.3 Street Scene dataset

The Street Scene dataset proposed by Ramachandra et al. [47] is designed for video anomaly detection and comprises 46 training and 35 testing high-resolution (1280×720) video sequences. These sequences were captured using a USB camera positioned to overlook a two-lane street with bike lanes and pedestrian sidewalks during the daytime. This dataset presents a challenging environment due to the diverse range of activities captured, including cars driving, turning, stopping, and parking; pedestrians walking, jogging, and pushing strollers; and bikers riding in bike lanes. Additionally, the videos feature changing shadows, moving backgrounds such as flags and trees blowing in the wind, and occlusions from trees and large vehicles.

The dataset includes 56,847 frames for training and 146,410 frames for testing, extracted at a rate of 15 frames per second. It contains a total of 205 naturally occurring anomalous events. These anomalies range from illegal activities like jaywalking and illegal U-turns to uncommon events not present in the training set, such as pets being walked and a meter maid issuing tickets. This variety makes the Street Scene dataset a comprehensive and demanding resource for advancing video anomaly detection research. However, a notable challenge of this dataset is that it is a single-scene dataset where only the Street Scene is included. This prevents models trained only on this dataset from generalizing their anomaly detection capabilities to other scenes. On the other hand, this dataset could be beneficial for models that are specifically designed to work on single scenes, such as EVAL [48]. The dataset is publicly available on: http://www.merl.com/demos/video-anomaly-detection.

5.2.4 UCF-Crime dataset

The UCF-Crime dataset [15] stands as a widely utilized large-scale dataset within recent research endeavors, boasting a multi-scene approach. Comprised of long, untrimmed surveillance videos, this dataset encapsulates 13 real-world anomalies with profound implications for public safety. The anomalies span a broad spectrum, including abuse, arrest, arson, assault, road accident, burglary, explosion, fighting, robbery, shooting, stealing, shoplifting, and vandalism.

To ensure the dataset's integrity and quality, a meticulous curation process was undertaken. Ten annotators were trained to collect videos from platforms like YouTube and LiveLeak, utilizing text search queries across multiple languages. Pruning conditions were enforced to exclude manually edited videos, prank videos, those captured by handheld cameras, and those sourced from news or compilations. The resultant dataset comprises 950 unedited real-world surveillance videos, each featuring clear anomalies, alongside an equal number of normal videos, totaling 1900 videos.

Temporal annotations were meticulously acquired by assigning the same videos to multiple annotators and averaging their annotations. The dataset is thoughtfully partitioned into a training set, consisting of 800 normal and 810 anomalous videos, and a testing set, comprising 150 normal and 140 anomalous videos. With its expansive coverage of various anomalous events, the UCF-Crime dataset serves as a comprehensive resource for evaluating anomaly detection algorithms across diverse real-world scenarios. The dataset is publicly available: http://crcv.ucf.edu/projects/real-world/.

5.2.5 CUHK Avenue

The CUHK Avenue dataset [19] is a widely used dataset for video anomaly detection, developed by researchers from the Chinese University of Hong Kong (CUHK). The dataset was recorded along the avenue of the CUHK campus. This dataset captures anomalies commonly seen in urban settings and public streets. It features a wide range of lighting conditions, weather changes, and human activities, making it a challenging benchmark for VAD models. Additionally, the dataset captures a wide range of anomalous behaviors, including physical actions like fighting and sudden running, as well as non-physical events such as unusual gatherings or people moving in the wrong direction. It also presents several challenges for VAD models, such as slight camera shake in some frames, missing examples of normal behavior in the training data, and the presence of outliers during training. In total, the dataset comprises 30,652 frames, with 15,328 frames allocated for training purposes and the remaining 15,324 frames for testing. With its diverse scenarios and realistic challenges, the CUHK Avenue dataset serves as a valuable benchmark for evaluating and advancing VAD techniques. The dataset is publicly available: http://www.cse.cuhk.edu.hk/leojia/projects/detectabnormal/dataset.html.

5.2.6 ShanghaiTech

The ShanghaiTech Campus dataset [18] is a major contribution to the field of anomaly detection, offering a large and diverse collection of video data. The dataset consists of 13 different scenes, with varying lighting conditions and camera angles, and it captures a wide range of real-world challenges. Notably, the dataset includes over 270,000 training frames, alongside 130 abnormal events, annotated at the pixel level to support detailed evaluation and analysis.

The dataset contains a total of 330 training videos and 107 testing videos, all consistently formatted at a resolution of 856×480 and a frame rate of 24 frames per second for smooth playback. Thanks to its large volume of data and detailed annotations, the ShanghaiTech Campus dataset has become a key resource for developing and evaluating anomaly detection methods in real-world environments. The dataset is publicly available: https://sviplab.github.io/dataset/campus_dataset.html.

5.2.7 XD-Violence

The XD-Violence dataset [17] is a large-scale and multi-scene dataset with a total duration of 217 h, containing 4754 untrimmed videos. This dataset comprises 2405 violent videos and 2349 non-violent videos, all of which include audio signals and weak labels. The focus of the dataset was on the field of weakly supervised violence detection, where only video-level labels are available in the training set. This approach offers the advantage of being more labor-saving compared to annotating frame-level labels. As a result, forming large-scale datasets of untrimmed videos and training a data-driven and practical system is no longer a challenging endeavor. The dataset incorporates audio signals and is sourced from both movies and in-the-wild scenarios. The dataset encompasses six physically violent classes: abuse, car accident, explosion, fighting, riot, and shooting. The dataset is divided into a training set, which comprises 3954 videos, and a test set, which includes 800 videos. Within the test set, there are 500 violent videos and 300 non-violent videos. https://roc-ng.github.io/XD-Violence/.

5.2.8 NWPU Campus dataset

The NWPU Campus dataset [49] is newly available. It represents a significant contribution to the field of video anomaly detection and anticipation. The dataset was compiled by setting up cameras at 43 outdoor locations on the campus to capture the activities of pedestrians and vehicles. To ensure a sufficient number of anomalous events, more than 30 volunteers were involved in performing both normal and abnormal activities. The dataset covers a wide range of normal events, including regular walking, cycling, driving, and other daily behaviors that adhere to rules. The anomalies encompass various categories such as single-person anomalies, interaction anomalies, group anomalies, scene-dependent anomalies, location anomalies, appearance anomalies, and trajectory anomalies. The dataset consists of 305 training videos and 242 testing videos, totaling 16 h of footage. Frame-level annotations are provided for the testing videos, indicating the presence or absence of anomalous events. Privacy concerns are addressed by blurring the faces of volunteers and pedestrians. Compared to other datasets, the NWPU Campus dataset stands out due to its larger scale, variety of scenes, and focus on scene-dependent anomalies. It is also the first dataset specifically designed for video anomaly anticipation. These qualities make it a valuable asset for pushing forward research in both anomaly detection and anticipation. The dataset is publicly available: https://campusvad.github.io/.

5.2.9 Discussion on datasets

The existing literature offers a wide range of datasets that capture both normal and abnormal scenarios across different contexts. Some focus on specific environments, like the UCSD Pedestrian dataset [46], which centers on a single scene, while others, such as UCF-Crime [15] and XD-Violence [17], cover multiple scenes and a broader variety of anomalies. These datasets also differ significantly in scale, with total durations ranging from just a few minutes to over 200 h of footage. Some datasets comprise a small number of long videos, primarily collected from fixed scenes, as seen in the subway dataset [45], while others contain a larger number of shorter videos sourced from a wide array of locations, like the XD-Violence dataset [17]. Additionally, the datasets showcase a noticeable diversity in the types of anomalies present, spanning from action-related anomalies such as unexpected movement direction and criminal activities to object-related anomalies like suspicious personnel and specific vehicle types.

Table 2 provides a comprehensive summary of the most commonly used publicly available datasets, while Fig. 4 illustrates sample frames extracted from these datasets.

Despite the availability of numerous datasets dedicated to the problem of VAD, several crucial trends and challenges need to be addressed:

Table 2 Overview of the most important publicly available datasets

Dataset	Balanced	# Videos	Length	Example anomalies	Frames/ video	Train:test	Challenges
Subway Entrance Gate [45] (2008)	No	1	1.5 hrs	Wrong direction, No payment	121,749	13:87	Single scene, indoor only, limited number of anomalies
Subway Exit Gate [45] (2008)	No	1	1.5 hrs	Wrong direction, No payment	64,901	13:87	Single scene, indoor only, limited number of anomalies
UCSD Ped1 [46] (2010)	No	70	5 min	Bikers, small carts, walking across walkways, wheelchair users	201	49:51	Small size, single scene, only outdoor, only vehicle anomalies
UCSD Ped2 [46] (2010)	No	28	5 min	Bikers, small carts, walking across walkways, wheelchair users	163	55:45	Small size, single scene, only outdoor, only vehicle anomalies
CUHK Avenue [19] (2013)	No	37	30 min	Strange action, Wrong direction, Abnormal object	839	50:50	Small size, single scene, outdoor only, camera shake
ShanghaiTech Campus [18] (2017)	No	437	3.67 hrs	Running, Loitering, Biking in Restricted Areas, Unusual Gatherings, Theft	1,040	86:14	Only university setting anomalies, single geographic location
UCF-Crime [15] (2018)	Yes	1,900	128 hrs	Abuse, arrest, arson, assault, accident, burglary, fighting, robbery	7,247	85:15	Imbalance between normal and abnormal classes, Variation in Video Quality
Street Scene [47] (2020)	No	81	226 mins	Jaywalking, illegal U-turns, pets, meter maid ticketing a car	2,509	57:43	Single geographic location, single scene, only outdoor
XD-Violence [17] (2020)	No	4,754	214 hrs	Abuse, Car Accident, Explosion, Fighting, Riot, Shooting	3,944	83:17	Limited number of anomalies, variation in video quality
NWPU Campus dataset [49] (2023)	No	547	16 hrs	Single-person, interaction, group, location, appearance, trajectory	2,527	56:44	Only university setting anomalies, single geographic location

- The majority of the publicly available VAD datasets offer limited environmental diversity and are restricted to a specific setting. Examples are the ShanghaiTech [18] and the CUHK Avenue [19] datasets, which are limited to university campus scenes and anomalies. Such a trend can significantly hinder the trained models' capability to generalize well in other types of settings.
- A recurring theme in multiple datasets is the limited number of anomalous events contained in the dataset. Some datasets contained as few as three anomalous event types (strange action, wrong direction, abnormal object, CUHK Avenue [19]) in comparison with other datasets that featured up to 11 types of anomalies (ShanghaiTech [18]). This constraint lessens the models' efficacy in practical applications by limiting their capacity to learn a broad range of potential abnormal behaviors.
- Most datasets available in the literature are imbalanced in terms of the normal and abnormal classes. Particularly in larger datasets like UCF-Crime [15], the imbalance between normal and abnormal classes makes it difficult for models to be trained to detect anomalies accurately without being surpassed by the majority class (normal).

We draw the following future directions for the VAD problem in surveillance videos, particularly in the realm of benchmark datasets for addressing the aforementioned issues:

- The benchmark datasets should be diverse and strive to encompass a wide range of anomalies, including both subtle and overt deviations from normal behavior.
- The datasets should aim for realism, mimicking real-world surveillance environments as closely as possible. Moreover, scalability is becoming important as surveillance video data continue to grow in size and complexity.

Fig. 4 Sample frames illustrating the variety of scenes and anomalies found in publicly available video anomaly detection datasets. These examples highlight the wide range of real-world scenarios and challenges that offer meaningful context for testing and benchmarking detection models

- Since many anomalies develop over time, capturing temporal context is essential. Therefore, benchmark datasets should incorporate long-term temporal information to represent how normal and abnormal behaviors evolve over time.
- Beyond detecting anomalies, it is important for future datasets to also focus on localizing and segmenting anomalous regions within video frames or sequences. This finer granularity improves the understanding of the nature and extent of anomalies, thereby enabling more prompt and targeted responses by the appropriate personnel.
- To mirror the complexity of real-world surveillance systems, future datasets could benefit from integrating data from multiple modalities, such as visual, auditory, and textual information, which can enhance the anomaly detection performance.
- Finally, openness and collaboration are key to advancing research in this field; therefore, future benchmark datasets should be openly accessible, inviting contributions and innovation from researchers worldwide in the field.

By focusing on these aspects, datasets can help drive the development of stronger and more reliable anomaly detection methods, ultimately making surveillance systems safer and more efficient.

5.3 Feature learning in deep video anomaly detection

Feature learning plays a pivotal role in the effectiveness of deep learning models for video anomaly detection. In this section, we take a closer look at the different feature extraction techniques utilized in the literature, emphasizing their significance and performance in surveillance video analysis.

5.3.1 Exploring different feature types

Video frames contain a variety of feature types that are important for video anomaly detection, including spatial, temporal, spatiotemporal, and even textual information.

Spatial features

Spatial features refer to the visual details found within each frame of a video. These include elements like shapes, textures, colors, and the positions of objects within the scene. In video anomaly detection, analyzing spatial features helps identify unusual patterns or objects that appear in certain areas of a video frame. Initially, traditional machine learning methods dominated this area, employing techniques like Gaussian mixture models and manually constructed features [21, 22]. However, the transition to deep learning facilitated automated feature extraction, thereby augmenting the capability to discern intricate spatial details within video data. Some recent works applied a hybridized approach that utilizes both classical image processing-based spatial features with deep learning models. A study by Sabzalian et al. [50] hybridized a deep CNN with histograms of optical flow (HOF) and histogram of oriented gradients (HOG) image features to boost the performance of VAD. Another study by Cheng et al. [51] proposed a hierarchical framework for VAD utilizing multi-level feature representation consisting of 3D scale invariant feature transform (3D-SIFT), HOF, and HOG, along with Gaussian process regression. However, classical image processing-based spatial features often struggle to capture the complex and high-level semantics present in video data, limiting their ability to detect subtle or context-dependent anomalies. In contrast, deep learning approaches excel at learning hierarchical and abstract representations, enabling them to adapt to diverse anomaly scenarios and generalize better across varying video contexts.

Temporal features

Temporal features capture changes or movements occurring over time in a video; this includes object motion, speed variations, and shifts in the environment from frame to frame. For instance, Liu et al. [52] employed the optical flow approach to track motion features across the frames. In video anomaly detection, temporal features play a pivotal role in identifying unusual actions or events in consecutive frames, such as unauthorized running in restricted areas [15, 20].

Another important aspect to consider in long videos is how to **sample temporal features or frames**, since processing every frame can be computationally intensive and may include redundant information. Researchers in VAD have utilized different sampling strategies to address this issue:

- **Uniform sampling:** This approach selects frames at regular intervals, regardless of content. Example: Sultani et al. [15] employ uniform sampling by dividing each video into non-overlapping segments and extracting C3D features from these segments. Specifically, they extract features from every 16 non-overlapping frames per video segment.
- **Sparse continuous sampling:** Apart from the traditional MIL-based approach that usually uses uniform sampling, sparse continuous sampling is proposed by Feng et al. [53] and also recently used in [54]. In this sampling strategy, video features are sampled into sub-bags consisting of t continuous snippets. The total number of features per video is then $M = \ell \times t$, where ℓ represents the number of sub-bags, and t is a hyperparameter that can be fine-tuned to capture the anomalous duration. This strategy reduces the complexity of the model while preserving key temporal contexts, leading to improved performance.

Spatiotemporal features

Relying solely on one type of feature can be restrictive: Temporal features might fail to pinpoint where an anomaly occurs, while spatial features might be overlooked when it happens. Incorporating both spatiotemporal features offers a more holistic perspective, capturing not only the occurrence but also the precise location of an anomaly. This approach leads to more accurate and effective anomaly detection [52, 55, 56].

Textual features

In video anomaly detection, textual features such as captioning and labeling significantly enhance the system's ability to recognize and understand anomalies. By incorporating descriptive captions and relevant labels into

video frames, these systems gain a deeper understanding of the content, context, and actions depicted in the videos. This semantic layer aids in distinguishing normal from abnormal activities more effectively. Advanced techniques like vision-language pretraining, including image-to-text, are employed to analyze these textual annotations, which can also encompass temporal and spatial context. The integration of textual features with visual data leads to more sophisticated, context-aware anomaly detection [28, 29, 31, 39].

5.3.2 Deep feature extractors and transfer learning

Previous researchers have employed different feature extractors in VAD, and transfer learning significantly enhances performance by utilizing pretrained models on large-scale datasets. This approach is particularly beneficial for VAD in scenarios with limited labeled data, as it reduces training time while improving detection accuracy.

Convolutional neural networks (CNNs)

- **2D Convolutional Neural Networks (2D CNNs):** CNNs have revolutionized spatial feature processing, enabling detailed analysis of structural elements in scenes. In the work of Mansour et al. [57], they discuss the use of Faster RCNN, a specific type of CNN architecture, because of its high accuracy and its dual capability of performing object classification and bounding box regression simultaneously. Transfer learning has proved to be a valuable approach in video anomaly detection. By fine-tuning pretrained Faster RCNN models, originally trained on large-scale datasets, researchers can adapt these models to effectively detect and localize anomalies within video frames.
- **3D Convolutional Networks (3D CNNs):** 3D CNNs have improved traditional CNNs by including temporal information, which makes them well-suited for capturing the spatiotemporal features of video data. 3D CNN models like C3D and I3D are pretrained on large action recognition datasets; they can also be fine-tuned for video anomaly detection tasks, which leads to significant performance gains. For example, Sultani et al. [15], Feng et al. [53], and Zhong et al. [37] have all employed the 3D CNN architectures as the backbone of their methods, which proves their effectiveness in capturing spatiotemporal patterns. In terms of results, Sultani et al. [15] achieved an AUC score of 75.41% on the UCF-Crime dataset, while Feng et al. [53] improved the performance to 82.30% and Zhong et al. [33] reported a frame-level AUC score of 82.12% on the same dataset, which demonstrates the steady progress and the growing impact of 3D CNNs in VAD research.

Autoencoders (AEs)

AEs are particularly effective in video anomaly detection because of their unsupervised learning ability (they can learn without needing labeled data). They work by compressing input data into a lower-dimensional space and then reconstructing it, learning key data features without needing labeled examples. This makes them especially useful for detecting anomalies, which are often rare, diverse, and not clearly defined.

In the study of Hasan et al. [20], the researchers utilized a fully convolutional autoencoder to learn low-level motion features. This helped the model to understand the regular motion patterns in long video sequences, making it well-suited for identifying irregular behavior. This method is flexible and can be used for different tasks, including temporal regularity analysis, frame prediction, and abnormal event detection in videos. Their model was able to achieve an AUC score of 81% on the UCSD Ped1 dataset and 90% on the UCSD Ped2 dataset, as well as 94.3% and 80.7% on the subway entrance and exit datasets, respectively.

Generative adversarial networks (GANs)

GANs consist of two main components: a generator that tries to create realistic data and a discriminator that tries to distinguish between generated and real data. Through this adversarial process, GANs effectively learn the distribution of the real data. This ability is particularly valuable in anomaly detection, as GANs can generate data that closely resembles normal instances, making it easier to identify anomalies that deviate from this learned pattern [25].

Their use in reconstruction-based approaches is especially noteworthy; these approaches reconstruct input data (like video snippets) using high-level representations learned from normal videos [58]. The premise is that anomalies, being out-of-distribution inputs, are harder to reconstruct accurately compared to normal data, making reconstruction error a viable metric for anomaly detection.

As seen in the work of Luo et al. [55], GANs are employed for future frame predictions, which are then reconstructed, showcasing their versatility and effectiveness in anomaly detection tasks. The authors achieved AUC scores of 85.7%, 84.3%, 96.2%, and 73.0% on the Avenue, UCSD Ped1, UCSD Ped2, and ShanghaiTech datasets, respectively.

In this domain, GANs alongside AEs are effective in capturing these intricate patterns in video data, aiding in the more precise identification of anomalous activities. Their combined use enables the learning of high-level representations and the generation of realistic data, enhancing the ability to detect anomalies accurately and efficiently.

Sequential deep learning

- **Long Short-Term Memory (LSTM)**: LSTMs are a special type of recurrent neural network designed to capture temporal dependencies. This characteristic makes them a great fit for working with sequential data, such as textual content in NLP-based systems [59]. In video applications like the VAD problem, their strength lies in the ability to remember and connect temporal patterns. This makes them especially useful for tracking and detecting unusual events that unfold over time in video streams [16].
- **Vision Transformers (ViT)**: Vision transformers have revolutionized the processing of sequential data by introducing attention mechanisms that allow models to focus on the most relevant parts of the input, which helps them better understand both temporal and spatiotemporal patterns. This makes ViTs especially effective at analyzing videos, where capturing both the appearance and movement over time is key to detecting complex anomalies. Yang et al. [60] introduce a novel video anomaly detection method by restoring events from keyframes, utilizing transfer learning on the pretrained swin transformer network (a special type of ViT). This approach, distinct from traditional methods, focuses on inferring missing frames and capturing complex spatiotemporal relationships, demonstrating improved anomaly detection in video surveillance. The authors were able to achieve an average AUC of 98.1%, 89.9%, and 73.8% on UCSD Ped2, Avenue, and ShanghaiTech datasets, respectively.

Vision-language models (VLM): Conventional techniques often rely solely on spatial–temporal characteristics, which might fall short in complex real-life situations where a deeper semantic insight is required for greater precision. Vision-language features involve training models using both visual and textual encoders, enabling comprehensive analysis of complex surveillance scenarios.

The notable rise of vision-language feature extractors using contrastive learning like the CLIP [38] and BLIP [61] leverages transfer learning by fine-tuning these pretrained models on vision-language tasks for anomaly detection, aligning vision and language data to improve the accuracy of detecting anomalies in complex environments. These models are designed to augment video content with rich semantic understanding, effectively narrowing the gap between simple pixel-based data and a more human-like interpretation of video content [39]. Additionally, the language models using text captions are used in text-to-video anomaly retrieval as in the work of Wu et al. [62]. Additionally, in their proposed VadCLIP methodology for video anomaly detection, Wu et al. [39] employed transfer learning by utilizing the pretrained CLIP model's vision-language features, which were integrated into a dual-branch system using both visual and textual encoders. It includes a local–global temporal adapter (LGT-Adapter) for modeling video temporal relations and leverages both visual and textual features for anomaly detection. One branch performs binary classification using visual features, while the other aligns visual and textual data for finer anomaly detection. In terms of the results, Wu et al. [39] achieved an excellent AUC score of 88.02% on the UCF-Crime dataset, outperforming several state-of-the-art models, and an 84.51% average precision score on the XD-Violence dataset. In comparison, Chen et al. [31] utilized transfer learning by fine-tuning the pretrained SwinBERT model to generate text features from video captions, enriching the semantic

understanding needed for detecting abnormalities. Their approach achieved a frame-level AUC of 98.7% on the UCSD Ped2 dataset, 98.1% on the ShanghaiTech dataset, 84.9% on the UCF-Crime dataset, and an average precision of 79.8% on the XD-Violence dataset.

These results highlight the promising potential of VLMs as powerful feature extractors, showing consistent improvements in anomaly detection performance across diverse datasets.

Graph-based features:

Graph neural networks (GNNs) are a type of neural network designed to handle data structured as graphs, where entities are represented as nodes and the relationships between them as edges [63, 64]. Many recent graph-based methods are inspired by graph signal processing theory [65–67]. GNNs are particularly effective at modeling complex relationships between data points by leveraging the structure of the graph itself. They operate using a mechanism known as message passing (or information diffusion), where each node updates its state by aggregating information from its neighboring nodes. This process allows GNNs to capture both local and global patterns within the graph. GNNs are suitable for both node-level tasks (such as node classification) and graph-level tasks (such as predicting properties of the entire graph) [68]. Over time, various GNN models have been developed, including graph convolutional networks (GCNs), which apply convolutional operations on graph data.

GNNs are commonly applied using one of the two main learning frameworks: **transductive** and **inductive** learning. **Transductive learning** relies on using the full dataset (including both training and test data) to model the relationships and propagate information across a graph. Although this approach can produce accurate predictions, it requires the entire graph to be retrained and regenerated whenever new data is introduced (such as when a new video is added in video anomaly detection). This makes transductive methods less scalable and efficient for dynamic datasets.

On the other hand, **inductive learning** allows models to generalize to unseen data without having to retrain on the whole dataset [69]. Instead of relying on a single large graph, multiple smaller graphs are constructed, which reduces the computational burden. This approach enables the model to make predictions on new data without fully rebuilding the graph, making it more adaptable and scalable.

In the VAD research, researchers have utilized the inductive learning approach while using GNNs.

For instance, Ullah et al. [70] proposed a weakly supervised GNN-assisted framework, called AD-Graph for VAD, where the GNN is used to model temporal and visual relationships between video segments. In this setup, they have treated each time segment as a node in the graph, and a graph convolutional layer is applied to these nodes to capture both motion and appearance information. By clustering similar nodes and applying spectral filters, AD-Graph leverages the structure of the graph to model the temporal dependencies and detect anomalies between the segments. The model achieved an AUC score of 75.2 % on the ShanghaiTech dataset and 83.85% on the UCF-Crime. A key contribution of another recently published approach (VadCLIP) [39] is the use of a GCN in their classification branch to model global temporal dependencies, identifying how video segments relate in terms of both feature similarity and temporal proximity.

Additionally, Zhong et al. [33] used GCNs to address the issue of label noise in weakly supervised video anomaly detection. They developed a method that constructs graphs based on feature similarity and temporal consistency, allowing high-confidence labels to propagate to low-confidence snippets. The authors obtained an excellent AUC score of 82.12% on UCF-Crime.

Hybrid feature learners:

Certain studies have developed hybrid feature extractors that integrate various types of feature extraction techniques to capture both spatial and temporal information effectively for video anomaly detection.

For example, Yuan et al. [71] proposed a hybrid architecture that combines U-Net and a modified video vision transformer (ViViT). The U-Net, known for its encoder-decoder structure with skip connections, excels at capturing detailed spatial information. In contrast, the modified ViViT, originally designed for video classification tasks, is adapted here to encode both spatial and temporal information effectively for video prediction. This hybrid model leverages the detailed high-resolution spatial information from the CNN features of the U-Net and

the global context captured by the transformer. Their model, which was proposed in 2021, outperformed the other SOTA on UCSD Ped1 and CUHK Avenue datasets with AUCs of 86.7% and 87%

Similarly, the work by Ullah et al. [10] employed a composite system that combines CNNs with transformers. Transfer learning is applied by using a pretrained EfficientNetV2 as the backbone for feature extraction.

In this configuration, the EfficientNetV2/CNN component is responsible for discerning spatial characteristics, whereas the transformers are tasked with identifying extended temporal attributes. This hybrid approach offers a complementary fusion of spatial and temporal features, enhancing the model's capability to detect anomalies in video sequences. Their model, which was proposed in 2023, outperformed the other SOTA on the ShanghaiTech, UCSD Ped2, and CUHK Avenue datasets with AUCs of 94.6%, 98.4%, and 89.6%.

Additionally, Nguyen et al. [72] proposed a CNN architecture that integrates convolutional autoencoders (CAEs) and a U-Net. Each stream within the network contributes uniquely to the task of detecting anomalous frames. The CAEs are responsible for extracting spatial features, while the U-Net focuses on capturing contextual information, enabling the model to effectively identify anomalies in video frames by leveraging both local and global features. Their model achieved a frame-level AUC of 86.9% and 96.2% on the CUHK Avenue and UCSD Ped2 datasets. Similarly, Majhi et al. [73] combined I3D 3D CNNs with LSTM networks to enhance anomaly detection performance. Transfer learning is applied by using a pretrained I3D CNN to extract spatiotemporal features from video frames, effectively identifying potential anomalies. However, as CNNs primarily capture short-term dynamics, LSTMs are integrated to capture long-range temporal dependencies crucial in lengthy, untrimmed surveillance footage. To handle the high-dimensional data from the I3D network, a pooling strategy is employed to optimize feature vectors for efficient processing by the LSTM. This hybrid architecture effectively integrates the strengths of both CNNs and LSTMs, enhancing the model's ability to detect anomalies in varying temporal scales. Their model achieved an AUC of 79.55% on the UCF-Crime dataset.

5.3.3 Future directions for feature learning

While each feature representation has its strengths and weaknesses, as empirically investigated by the SOTA VAD approaches [10, 30, 39, 60, 74], the end-to-end deep feature learning paradigms are the recent trends, and these approaches have advanced the VAD performance. We provide the following future directions in terms of the feature representations to improve detection, accuracy, and efficiency.

- A continuous exploration and refinement of deep learning architectures tailored for VAD are needed. This may involve developing more efficient architectures, such as spatiotemporal CNNs, ViTs, or RNNs, that can effectively capture temporal dependencies and spatial information in video sequences.
- The hybrid models, which integrate multiple types of features, including appearance, motion, and spatiotemporal information, could also improve the performance. This could involve combining deep learning-based features with handcrafted features or leveraging multimodal approaches such as VLMs to capture diverse aspects of anomalies. The hybridization could be performed for fusing information at different scales. An excellent example of that is the investigation of fusing analyzed local temporal activities with global video-level context information. This allows the model to look at the general context when deciding if a given video snippet is anomalous or not.
- Investigation of cross-modal feature representations that integrate information from multiple modalities, such as visual, audio, and textual cues present in surveillance videos. Cross-modal representations can capture complementary information sources and improve the robustness of anomaly detection models to diverse types of anomalies and environmental conditions. An excellent example of that is the HL-Net [17] model, which fuses information from visual and audio inputs, which helped in enhancing the performance, especially in the cases of anomalies that are difficult to analyze using the appearance alone.
- Exploration of self-supervised learning techniques for pretraining feature representations in an unsupervised manner can help in learning rich representations from unlabeled data. The pretrained models can then be fine-tuned for specific anomaly detection tasks with limited labeled data. Analyzing the use of different pretext

tasks optimized for maximizing learning from unlabeled video footage can help in boosting the performance of VAD models, particularly under limited supervision.

- Integration of attention mechanisms into deep learning architectures to focus on relevant spatiotemporal regions or frames within videos. Attention mechanisms can help improve the model's ability to attend to informative parts of the input data and suppress irrelevant noise, leading to more robust anomaly detection features. An example of that is the Florence-VL [75] model, which utilizes an attention mechanism to fuse textual and visual information for enhancing VAD performance. More work could investigate different uses of attention-based architectures for extracting different types of information from videos.
- Utilization of graph-based feature representations to model complex relationships among different entities (e.g., objects, regions) within surveillance videos. GNNs can capture dependencies and interactions among entities, enabling more effective anomaly detection in scenarios where anomalies manifest as abnormal interactions or relationships. Novel applications of recent graph-based architectures such as VCR-Graphormer [76] could significantly enhance the models' ability to extract important features for VAD.
- Investigation of adversarial learning techniques to enhance the robustness of feature representations against adversarial attacks. Adversarial training methods can help improve the model's ability to generalize to unseen anomalies and mitigate the risk of evasion attacks in real-world surveillance systems. This is of extreme importance in cases where the surveillance data are highly confidential, pushing the methods closer to practical application.
- Development of incremental learning approaches to adaptively update feature representations over time as new data become available can help the model adapt to evolving patterns of normal and abnormal behavior in dynamic surveillance environments without forgetting previously learned knowledge. This makes the model very suitable for practical application, as more scene-specific data can result in a notable performance improvement.

5.4 Learning and supervision schemes

In the context of this survey, we will delve into various learning approaches aimed at addressing the VAD problem. In Fig. 5 we illustrate various approaches to VAD, including supervised, semi-supervised, unsupervised, self-supervised, and weakly supervised methods. In the supervised setup, both training and testing involve frame-level labels (Y_{v_i}), distinguishing between normal (N) and anomalous (A) frames.

In the semi-supervised setup, only normal data (N) are used during training, while testing involves both normal and anomalous data. This paradigm is normally solved by the literature using reconstruction-based or prediction-based methods, where the model is trained to effectively represent normal events, and anomalies are detected

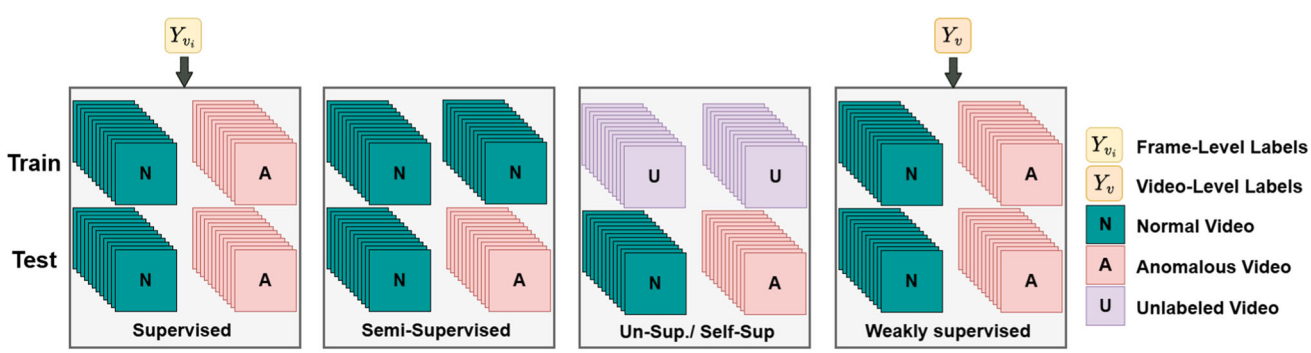

Fig. 5 Comparisons of different supervision schemes in VAD tasks (supervised, semi-supervised, unsupervised, self-supervised, and weakly supervised)

based on poor reconstruction, as they deviate from the learned normal behavior. These methods are particularly useful when labeled anomaly data is scarce.

The self-supervised methods also rely on unlabeled (U) data; they rely on auxiliary or proxy tasks to extract supervisory signals directly from unannotated data. In essence, it does not require external labeled data, as the labels are generated internally from the input data itself. In the context of semi-supervised VAD, where explicit supervisory signals are absent, self-supervised learning is crucial for learning normal patterns and building a normal representation using these auxiliary tasks. As a result, self-supervised methods consistently lead to semi-supervised VAD.

The unsupervised methods rely on fully unlabeled (U) data for training, with no manual labels provided. These methods discover anomalies without any explicit supervision, often using generative or cooperative learning approaches.

In the weakly supervised approach, training and testing involve both normal (N) and anomalous (A) videos, but only video-level labels (Y_v) are available. The model is tasked with classifying entire videos as normal or anomalous without frame-level annotations, requiring it to determine which specific regions or frames within the video are abnormal.

The figure demonstrates how these supervision schemes vary by the level of dataset annotation—frame level versus video level—which influences the choice of loss functions. For instance, frame-level annotations typically use classification losses (e.g., cross-entropy), video-level annotations suit multiple instance learning loss, and semi-supervised approaches often employ reconstruction loss to model normal behavior for anomaly detection. Further details on these losses are discussed in Sect. 5.4

5.4.1 Supervised approaches

In supervised learning, algorithms are developed using datasets where each frame is prelabeled as either "normal" or "anomalous." This allows the model to learn explicit distinctions between normal and abnormal events based on these annotations. However, the use of supervised methods in VAD is relatively rare. This is primarily because acquiring datasets with detailed, frame-level annotations is highly challenging and resource-intensive. Anomalous events are often rare and vary widely in nature, which makes it challenging to build a comprehensive set of labeled data. Additionally, manually labeling each frame in extensive video data is time-consuming and resource-intensive. While supervised learning can be powerful when precise labels are available, its application in VAD is constrained by the practical challenges in obtaining extensive and accurately annotated training data.

A study by Zhou et al. [77] presented a method for detecting and localizing anomalies in crowded scenes using spatial–temporal convolutional neural networks (CNNs). Their method can process both spatial and temporal data from video sequences, allowing it to capture appearance and motion information. Additionally, it identifies anomalies by focusing on moving pixels, leading to enhanced accuracy and robustness. Similarly, Chen et al. [58] addressed the scarcity of labeled anomalies in anomaly detection by employing a conditional generative adversarial network (cGAN). Their method generates additional training data that is class-balanced. This approach utilizes labeled data for training the model. They propose a novel supervised anomaly detector, the ensemble active learning generative adversarial network (EAL-GAN). This network, unique in its architecture of one generator against multiple discriminators, leverages an ensemble learning loss function and an active learning algorithm. The goal is to alleviate the class imbalance problem and reduce the cost of labeling real-world data.

5.4.2 Self-supervised approaches

Self-supervised approaches involve training models using data that are not explicitly labeled for anomalies. Instead, the models learn to identify anomalous events by solving "proxy tasks," which generate supervisory signals from the data itself. These tasks are designed to be related to the main goal of detecting anomalies, helping the model to develop an understanding of normal patterns in the data without needing direct supervision from

labeled examples of anomalous events. However, it demands careful design and selection of proxy tasks to ensure that the learned features are useful for identifying anomalous events. In the work by Georgescu et al. [13], researchers propose a method based on self-supervised and multitask learning at the object level. Self-supervision involves training a 3D CNN on several proxy tasks that do not require labeled anomaly data. These tasks include determining the direction of object movement (arrow of time), identifying motion irregularities by comparing objects in consecutive versus intermittent frames, and reconstructing object appearances based on their preceding and succeeding frames. By learning from the video data itself what normal object behavior looks like, the model becomes adept at detecting anomalies when deviations from this learned behavior occur. This methodology enables effective anomaly detection even in the absence of explicit labels.

Their approach achieved frame-level AUC scores of 92.8% on Avenue and 99.8% on UCSD Ped2 datasets and a macro AUC of 89.3% on ShanghaiTech. The authors continued their work in [78]. The new enhancements include integrating advanced object detection methods such as YOLOv5, optical flow, and background subtraction, which improve the detection of rapidly moving objects and those outside predefined classes. They also introduce transformer blocks into the architecture, exploring 2D and 3D convolutional vision transformers (CvT) to better capture complex spatiotemporal dependencies. These updates culminate in a more robust framework that significantly enhances the detection accuracy and adaptability in identifying anomalous events in video sequences. Their new approach significantly boosted performance, achieving 93.7% micro AUC on the Avenue dataset and 90.5% macro AUC on the ShanghaiTech dataset.

5.4.3 Semi-supervised approaches

The semi-supervised approaches rely only on normal data during training, while testing involves both normal and anomalous data. This paradigm is normally solved by the literature using reconstruction-based or prediction-based methods

Reconstruction-based

Reconstruction-based approaches for video anomaly detection operate under the principle that normal events can be effectively reconstructed from a learned representation, while anomalies or abnormal events deviate significantly from this representation and are harder to reconstruct. Essentially, the model learns to represent or "reconstruct" normal data, and anomalies are detected based on how poorly the model reconstructs them. In certain studies, the reconstruction-based approaches are referred to as the one-class classification (OCC) paradigm, where the model is trained only on data from one class (typically the "normal" class), and any deviation from this learned structure in the testing phase "anomaly" results in higher reconstruction error.

These methods are particularly suitable when anomaly data are scarcely labeled. During training, only normal videos are considered, while during testing, the model is evaluated on both normal and abnormal videos to assess its anomaly detection capabilities. Although these models are feasible, scalable, and cost-effective approaches. However, the effectiveness of these models is highly dependent on the quality and comprehensiveness of the normal training data. If the normal data are not representative of all possible normal variations, the model's performance in detecting anomalies can suffer.

Another issue is that the model could generate a high number of false positives, labeling normal variations or benign deviations as anomalies. This occurs because any deviation from the learned normal patterns, even if it is not truly anomalous, might be flagged.

Deep learning techniques, especially CNN [72] or autoencoders [20, 79], are widely utilized for this approach. An autoencoder attempts to learn a compressed representation of its input data and then reconstructs the original data from this representation. During training, the model learns to minimize the reconstruction error between the input and the output. After training, when the model encounters new data, it tries to reconstruct it based on what it has learned. The difference between the reconstructed output and the original input is measured, typically as a reconstruction error. A high reconstruction error indicates that the input is significantly different from what the model considers "normal" implying that the input might be an anomaly.

In the work by Hasan et al. [20], the authors developed a method to assess the regularity of frames in video sequences using reconstruction models. They employed two autoencoders: one with convolutional layers and the other without. The model processed two distinct types of inputs: manually designed features (such as HOG and HOF, enhanced with trajectory-based characteristics), and a combination of 10 successive frames aligned along the time axis. The error in reconstructing these frames is often used as an indicator of their regularity score.

The evolution of the reconstruction approaches to future frame prediction:

Over time, the idea of using reconstruction-based methods in video anomaly detection has evolved. Researchers observed that deep neural networks may not always produce significant reconstruction errors for unusual events. Additionally, for some normal videos that are previously unseen, they could be incorrectly labeled as abnormal. To overcome this, researchers shift their focus toward the prediction of future frames based on previous video frames by considering optical flow constraints to ensure motion consistency. This evolved methodology, termed as *prediction approaches*, introduced generative adversarial networks (GANs), which played a significant role in enhancing this approach [52, 55]. Building upon this framework, the HSTGCNN [80] model incorporates a sophisticated future frame prediction (FFP) mechanism that significantly refines the anomaly detection process. By integrating hierarchical graph representations, the model not only predicts future frames but also encodes complex interactions among individuals and their movements, thus providing a more robust and context-aware anomaly detection system.

A hybrid system incorporating flow reconstruction and frame prediction was introduced by Liu et al. [24]. This system detects anomalous events in videos by learning normal activity patterns and predicting future frames based on those learned features. It utilizes a memory-augmented autoencoder for accurate flow reconstruction and a conditional variational autoencoder for frame prediction. Anomalies are highlighted by larger errors in flow reconstruction and subsequent frame prediction.

5.4.4 Unsupervised approaches

Unsupervised learning aims to understand the underlying structure or distribution of the data without requiring labeled examples. It leverages unlabeled data for training, learning to reconstruct the data and capture inherent patterns without explicit labels indicating what is normal or anomalous.

As the work of Zaheer et al. [81], proposed a novel approach called generative cooperative learning, which combines a generator and a discriminator trained together using a negative learning paradigm. The generator, designed as an autoencoder, reconstructs both normal and abnormal representations. Through negative learning, the discriminator learns to estimate the probability of an instance being abnormal by identifying the reconstruction errors from the generator. This method leverages the assumption that anomalies are less frequent than normal events and that normal events exhibit temporal consistency, allowing for more effective anomaly detection.

Another unsupervised approach by Tur et al. [82] utilizes diffusion models, a type of generative model, to leverage the reconstruction capabilities for detecting anomalies. The approach begins by extracting features from video clips using a 3D convolutional neural network (3D CNN). These features are then fed into a diffusion model, which reconstructs the features without relying on labeled data. The diffusion model progressively adds Gaussian noise to the input data and learns to reverse this process, effectively reconstructing the input.

5.4.5 Weakly supervised approaches

In weakly supervised approaches, unlike supervised approaches, obtaining precise frame-level annotations for anomalies in long video sequences can be challenging and time-consuming. Instead, annotators may label "snippets" or short segments within the video where anomalies are observed, serving as weak labels for training the model.

The concept of weak supervision was first introduced by Sultani et al. [15] through their pioneering multiple instance learning model. This model was the first to adopt the notion of weakly labeled training videos. In this approach, normal and anomalous videos are considered negative and positive bags, respectively, with video segments serving as instances in MIL. These bags are processed through feature extractors to capture spatiotemporal features before being directed to a fully connected network to produce the final output. The output anomaly score ranges from 0 to 1, to increase the model's output anomaly score for abnormal segments while decreasing it for normal segments. However, this method can introduce noisy labels since the weak annotations do not provide precise information about the exact location of anomalies within the segments. This ambiguity can lead to the model learning less accurate representations of normal and abnormal behaviors. Additionally, the reliance on segment-level labels can sometimes result in the inclusion of irrelevant frames within the labeled segments, further complicating the training process.

Other works have solved this issue. For example, the authors Zhong et al. [33] introduced a novel approach to weakly supervised anomaly detection (WSVAD), reframing it as a supervised learning task with noisy labels, departing from the conventional MIL framework. Leveraging a graph convolutional network (GCN), this method effectively cleans noisy labels, thereby enhancing the training and reliability of fully supervised action classifiers for anomaly detection. Additionally, the authors Yang et al. [83] introduced a binarization-embedded WSVAD (BE-WSVAD) method, which innovates by embedding binarization into the GCN-based anomaly detection module.

Traditional MIL approaches often neglect the intricate interplay of features over time. A study by Zhang et al. [32] improves traditional MIL with a temporal convolutional network (TCN) and a unique inner bag loss (IBL). The IBL strategically focuses on the variation of anomaly scores within each bag (video), emphasizing a larger score gap in positive bags (containing anomalies) and a smaller gap in negative bags (without anomalies). Meanwhile, the TCN effectively captures the temporal dynamics in videos, a crucial aspect often overlooked in standard MIL approaches.

Moreover, the authors Zaheer et al. [34, 84] propose a self-reasoning approach based on binary clustering of spatiotemporal video features to mitigate label noise in anomalous videos. Their framework involves generating pseudo-labels through clustering, facilitating the cleaning of noise from the labels, and enhancing the overall anomaly detection performance. This method not only removes noisy labels but also improves the network's performance through a clustering distance loss.

Purwanto et al. [36] proposed a method that commences with a relation-aware feature extractor that captures multi-scale CNN features from videos. The unique aspect of their approach is the integration of self-attention with conditional random fields (CRFs), leveraging self-attention to capture short-range feature correlations and CRFs to learn feature interdependencies. This approach offers an analysis of complex movements and interactions for anomaly detection. On the other hand, Tian et al. [37] proposed robust temporal feature magnitude learning (RTFM) to tackle the challenge of recognizing rare abnormal segments in videos that are mostly composed of normal activity. In particular, this is useful for detecting subtle anomalies that differ only slightly from the normal behavior. RTFM employs temporal feature magnitude learning to improve the robustness of the MIL approach against negative instances of abnormal videos; it also integrates dilated convolutions and self-attention mechanisms to capture both long- and short-range temporal dependencies.

In a related work, Feng et al. [53] introduced "MIST: Multiple Instance Self-Training Framework for Video Anomaly Detection," a novel method for WSVAD. Unlike traditional MIL approaches, MIST introduces a pseudo-label generator with a sparse continuous sampling strategy for more accurate clip-level pseudo-labels. It also includes a self-guided attention-boosted feature encoder for focusing on anomalous regions within frames.

Additionally, Zhang et al. [85] present a novel weakly supervised temporal relation learning framework (WSTR). The framework, which uses I3D for feature extraction and incorporates snippet-level classifiers and top-k video classification for weak supervision, is the first of its kind to apply transformer technology in this context.

Joo et al. [30] proposed a novel approach called CLIP-TSA that leverages vision-language (ViT) features for WSVAD. Unlike conventional models such as C3D or I3D, CLIP-TSA utilizes the vision transformer (ViT)

encoded visual features from CLIP [38] to efficiently extract discriminative representations. It incorporates a temporal self-attention (TSA) mechanism to model both long- and short-range temporal dependencies, thereby enhancing detection performance in VAD.

Another recently published work by [74] proposes a novel weakly supervised framework called text prompt with normality guidance (TPWNG), leveraging the CLIP model for aligning textual descriptions with video frames to generate pseudo-labels. The method involves fine-tuning CLIP for domain adaptation using ranking and distributional inconsistency losses and introducing a learnable text prompt mechanism with normality visual prompts to improve text-video alignment. The framework also incorporates a pseudo-label generation module based on normality guidance to infer reliable frame-level pseudo-labels and a temporal context self-adaptive learning module to flexibly capture temporal dependencies in video events. Finally, the most recent approach, [39], achieves the highest performance on two public datasets, similar to the TPWNG work; the authors utilized the vision and text encoders from the CLIP model. However, VadCLIP uses a dual-branch architecture. One branch uses visual features for coarse-grained binary classification. The other leverages fine-grained language-image alignment. It focuses on transferring the pretrained knowledge from CLIP to the WSVAD task without altering the model.

5.4.6 Future directions for learning and supervision schemes

Different VAD paradigms have their strengths and weaknesses. ViT-based approaches with the integration of language models/language supervision are the recent trends for improving VAD performance. In the realm of different approaches for VAD, several promising directions are emerging:

- Development of vision-language models for VAD problems will be more dominant compared to vision-only models. Examples of such vision-language models include ALBEF [86], ALPRO [87], BLIP-2 [61], and PNP-VQA [88]. They are capable of handling various tasks, including visual question answering (VQA) [89], visual reasoning [90], image and video captioning [91], and visual grounding [92]. These models not only capture the semantic representation of the videos but also consider the natural language descriptions for anomalies. This allows the models to combine visual cues and language semantics, which helps them discover deeper patterns in the video footage.
- Exploration of methods to localize and describe anomalies in videos using natural language descriptions is the recent trend. Language-guided anomaly localization techniques enable the model to identify spatial and temporal regions corresponding to anomalies and generate human-interpretable descriptions, enhancing situational awareness and facilitating response efforts. An example of this is the Holmes-VAU [93] model, which provides an explanatory text generated using a multimodal large language model (LLM), allowing both for human-understandable decisions and improved temporal anomaly localization capability.
- Exploration of GAN-based approaches for VAD, where the generator learns to generate normal video frames while the discriminator distinguishes between real and generated frames. GANs can capture complex distributions of normal behavior and detect deviations indicative of anomalies. A recent application of this idea is the CGL [81] model, which used a GAN-based architecture for improving unsupervised VAD performance. Similar works could focus on novel applications of similar architectures for performing VAD under different supervision conditions, such as semi-supervised and weakly supervised VAD.
- Development of graph-based models that represent spatial and temporal dependencies among entities in surveillance videos as a graph structure. Spatiotemporal graph models can capture intricate relationships among objects, activities, and contexts, enabling more accurate anomaly detection in complex scenes. An excellent application of this type of architecture for enhancing VAD performance is the VadCLIP [39] model, which utilizes a local–global temporal (LGT) adapter that includes a graph convolutional network for extracting the global temporal patterns for a sequence of snippet features. Recent graph-based neural network architectures such as the Graphormer [94] and graph isomorphism network (GIN) [95] could be utilized for learning more complex spatiotemporal relationships in video data for enhancing VAD performance.

- Investigation of transfer learning techniques to transfer knowledge from labeled source domains to unlabeled or sparsely labeled target domains can help mitigate the scarcity of labeled data in target domains and improve anomaly detection performance by leveraging knowledge learned from related surveillance scenarios. This technique is significant because most VAD datasets are either unlabeled or weakly labeled at a video level, which is much coarser than the required frame level. Leveraging higher-quality labeled sources can therefore boost performance in scenarios where training data is insufficient.
- Adoption of multi-resolution analysis techniques to analyze videos at different spatial and temporal resolutions. Multi-resolution approaches enable the detection of anomalies occurring at varying spatiotemporal scales, from small-scale events to large-scale spatial or temporal patterns, enhancing the model's sensitivity to diverse types of anomalies. This is of crucial importance as different types of anomalies could vary significantly in both their visual appearance and duration, where some might be small or quick while others are large or prolonged, thus having robust methods that consider both is required.
- Adoption of dynamic ensemble learning strategies to combine predictions from multiple anomaly detection models dynamically. These multiple anomaly detection models could consist of different feature extractors, model architectures, or processed spatiotemporal scales. Dynamic ensemble methods can adaptively adjust the ensemble composition based on the current surveillance context, improving detection robustness and resilience to evolving anomalies.

5.5 Loss functions

Loss functions play a pivotal role in quantifying the disparity between predicted outcomes by a model and the actual results. They serve as guides for the optimization process, facilitating adjustments and enhancements to model parameters during training. The selection of an appropriate loss function is critical as it directly impacts the model's capacity to discern underlying patterns in the data and, consequently, its performance on unseen data. Various tasks necessitate distinct loss functions to adeptly capture the specific characteristics or complexities inherent in the problem domain.

5.5.1 Multiple instance learning (MIL) loss

Within the category of weakly supervised learning, notably multiple instance learning (MIL) [15], the objective function is crafted to proficiently distinguish between normalcy and anomaly within video segments.

When dealing with a dataset of labeled videos, denoted as $\mathcal{V}$, where V is a video segment, the videos are categorized into positive bags $\mathcal{B}_+$ and negative bags $\mathcal{B}_-$. The overall objective function is represented as,

$$\arg\min_{\theta} \sum_{\mathcal{B}_+,\mathcal{B}_-\in\mathcal{V}} \max\left(0, 1 - \max_{V\in\mathcal{B}_+} S(V;\theta) + \max_{V\in\mathcal{B}_-} S(V;\theta)\right) \tag{2}$$

This function aims to ensure that the anomaly score $S(V;\theta)$ for the positively labeled video dataset is greater than any segment within the negatively labeled set. This is achieved by employing a hinge loss function which is parameterized by θ.

5.5.2 Cross-entropy loss

- Alongside the MIL loss, some other researchers like Zhong et al. [33] and Lv et al. [96] used the binary cross-entropy within their loss functions to classify between normal and abnormal video snippets in the WSVAD tasks.

  The goal is to refine the model to identify the snippet with the highest anomaly score in a video, which is

denoted as normal, and conversely, in an anomalous video. For each video instance, MIL formulates a pair that includes the model's prediction for the snippet with the highest anomaly score and the true label for the video's anomaly (e.g., $\max\{f(V_i)\}_{i=1}^{n}, y_i$), where $y_i = 0$ for normal and $y_i = 1$ for anomalous instances). MIL subsequently consolidates these pairs from all the videos to form a set C of confidently labeled snippets. The model f is refined by optimizing the binary cross-entropy (BCE) loss, calculated as follows:

$$BCE(C) = -\frac{1}{N}\sum_{(y_i,\hat{y}_i)\in C}\left[y_i \log(\hat{y}_i) + (1-y_i)\log(1-\hat{y}_i)\right] \tag{3}$$

where the maximum score $\hat{y}_i = \max\{f(V_i)\}_{i=1}^{n}$. Under this paradigm, the model f must assign the lowest anomaly probability to all snippets in a normal video, thereby minimizing $\max\{f(V_i)\}_{i=1}^{n}$, and the highest anomaly probability in an anomalous video, hence maximizing $\max\{f(V_i)\}_{i=1}^{n}$. The strategy is to focus on the snippet with the strongest abnormality, even if the video is largely normal, to generate a set of snippets labeled with high confidence as anomalous.

- In WSVAD, the presence of noisy or incomplete labels poses significant challenges. To address this, uncertainty-aware loss functions incorporate measures of uncertainty into the training process, allowing the model to weigh each sample's contribution based on the confidence in its pseudo-label. This approach enhances the model by focusing more on reliable samples.

  For instance, Zhang et al. [97] using the BCE loss, proposed an uncertainty-aware loss that utilizes Monte Carlo dropout [98] to estimate the uncertainty of pseudo-labels in weakly supervised VAD. Specifically, they perform M stochastic forward passes through the model with dropout enabled, obtaining multiple predictions $\hat{Y}_i^{(m)}$ for each sample x_i. The predictive mean of the pseudo-labels is calculated as:

$$\mathbb{E}[\hat{Y}_i] = \frac{1}{M}\sum_{m=1}^{M}\hat{Y}_i^{(m)} \tag{4}$$

  and the predictive uncertainty (variance) is estimated as:

$$\mathrm{Var}[\hat{Y}_i] = \frac{1}{M}\sum_{m=1}^{M}\left(\hat{Y}_i^{(m)}\right)^2 - \left(\mathbb{E}[\hat{Y}_i]\right)^2 \tag{5}$$

  Samples with lower uncertainty are considered more reliable. The model selects a subset of samples Ω_R with the lowest uncertainties to form a reliable set for training. The BCE loss is then used to train the clip classifier f_c on these reliable samples:

$$\mathcal{L}_{\mathrm{BCE}} = -\sum_{(x_i,\tilde{y}_i)\in\Omega_R}\left[\tilde{y}_i \log f_c(x_i) + (1-\tilde{y}_i)\log(1-f_c(x_i))\right] \tag{6}$$

  where $\tilde{y}_i = \mathbb{E}[\hat{Y}_i]$ is the refined pseudo-label for sample x_i. By iteratively training the model on these low-uncertainty samples, the approach refines the pseudo-labels and improves the model's robustness and accuracy in detecting anomalies.

5.5.3 Reconstruction error loss

For the reconstruction-based approaches, the objective function of the model, as shown in Eq. 7, is to minimize the reconstruction error (the squared Euclidean distance, $\|\cdot\|_2^2$), as proposed by Hasan et al. [20], between the original input frames pixel values and its reconstruction by the model with weights θ.

The input frames denoted as V_i are passed through an encoder to obtain a compressed representation, which is then reconstructed back into frames $F(V_i;\theta)$ using a decoder, and N is the size of the minibatch.

$$\arg\min_{\theta} \frac{1}{2N} \sum_{i} \|V_i - F(V_i;\theta)\|_2^2 \tag{7}$$

5.5.4 Adversarial loss

Adversarial loss is a key component in the GAN-based approaches. To model the distribution of normal video VAD tasks, adversarial training between a generator G and a discriminator D is employed. The generator G is trained to map input samples back to the normal data distribution, while the discriminator D aims to distinguish between genuine normal samples and those produced by G. The adversarial loss function, as proposed by Sabokrou et al. [99], guiding this training process is defined as in Eq. 8:

$$\min_{G} \max_{D} \left\{ \mathbb{E}_{x\sim p}[\log D(x)] + \mathbb{E}_{\tilde{x}\sim p+\mathcal{N}(0,\sigma^2)}[\log(1 - D(G(\tilde{x}_i)))] \right\} \tag{8}$$

In this equation, x represents samples drawn from the normal data distribution p, and $\hat{x} = x + \epsilon$ are these samples with added Gaussian noise $\epsilon \sim \mathcal{N}(0, \sigma^2)$ with mean 0 and variance σ^2. The discriminator D outputs the probability that an input is a real normal sample, while the generator G attempts to produce outputs that D cannot distinguish from real data. The anomaly score for a given input x is calculated as $D(G(x))$, where a lower score indicates a higher likelihood of the sample being anomalous.

5.5.5 Future directions for loss functions

Future directions for video anomaly detection in terms of loss functions involve the exploration and development of novel loss functions tailored to address specific challenges and objectives in anomaly detection tasks. The promising future directions in this domain are:

- Designing anomaly-aware loss that explicitly accounts for the characteristics of anomalies could penalize model errors differently based on the severity or rarity of anomalies, helping the model focus more on detecting critical anomalies while reducing false alarms for common or benign events. Examples of such anomalies are focal loss [100] and affinity loss [101]. These could significantly help mitigate class imbalance problems and harness context and locality in video data.
- The development of constraints aiming to guarantee temporal consistency in the model predictions is required to ensure coherent anomaly score predictions.
- The addition of adversarial losses to VAD training losses is of critical importance, especially for developing robust and practical surveillance systems that can withstand different types of adversarial attacks in the input data.
- Incorporating uncertainty-aware losses to quantify and mitigate uncertainty in anomaly detection predictions may enable the model to estimate the confidence or uncertainty associated with its predictions, facilitating more reliable decision-making and uncertainty quantification. These losses penalize overconfident predictions on uncertain or ambiguous events while providing better generalization to unseen events.

5.6 Regularization

Regularization techniques are important for mitigating the problem of overfitting and enhancing VAD models' generalization to different settings.

5.6.1 Weight decay regularization

Weight decay is a common type of regularization in which the weights' size of the trainable model components is reduced throughout the training process. This is performed by adding an additional term to the loss function, which penalizes the overall size of the model's weights [102]. Guaranteeing a small size of the model's weights reduces the overall complexity, thus improving the generalizability by preventing the model from relying solely on a specific subset of the input features.

L1 regularization

L1 regularization penalizes the weights' size by adding the absolute value of the magnitude of the weights of the model as an additional term in the loss function. The general equation for a loss function with L1 regularization can be expressed generally as follows:

$$L(\theta) = L_0(\theta) + \lambda \sum_{i=1}^{n} |\theta_i| \tag{9}$$

where $L(\theta)$ is the total loss function after including the L1 regularization term, $L_0(\theta)$ is the original loss function without regularization, and λ is a regularization parameter that controls the strength of the regularization effect. Higher values of λ lead to greater regularization, potentially resulting in more features being effectively ignored by reducing their coefficients to zero. The key aspect of L1 regularization is its ability to create sparse models. Sparsity here refers to the property where some of the coefficients become exactly zero, effectively excluding some features from the model. This is particularly useful in feature selection, where the most important features in the model are to be identified.

L2 regularization

L2 regularization, also known as ridge regularization, is a technique that modifies the loss function of a model by adding a penalty equivalent to the square of the magnitude of the model's weights. The formula for a loss function with L2 regularization is generally expressed as follows:

$$L(\theta) = L_0(\theta) + \lambda \sum_{i=1}^{n} \theta_i^2 \tag{10}$$

Here, $L(\theta)$ represents the total loss function, including the L2 regularization term. $L_0(\theta)$ is the original loss function without regularization, and λ is the regularization parameter, which determines the strength of the regularization effect. Increasing the value of λ enhances the regularization effect, penalizing larger weights more significantly. The key characteristic of L2 regularization is its ability to prevent overfitting by keeping the weights small, which leads to a more generalized model. Unlike L1 regularization, which promotes sparsity in the model (some coefficients becoming exactly zero), L2 regularization tends to shrink all coefficients toward zero but does not make them exactly zero. This attribute makes it particularly useful in scenarios where many features contribute small amounts to the predictions. By penalizing the square of the weights, L2 regularization ensures that no single feature dominates the model, which can be essential for models where all features carry some level of importance.

5.6.2 Temporal and spatial constraints

Temporal and spatial constraints are other types of regularization terms that are commonly used in VAD [15]. The addition of these terms in the loss function ensures important specifications in the model's spatial and temporal learning for distinguishing between the normal and abnormal events in videos [103]. Temporal constraints are used to guarantee the continuity and progression of events over time, while spatial constraints are used to enhance the model's understanding of the objects' positions and physical locations throughout the frames of the video and their relation to the existence of anomalies. By incorporating both types of constraints in the loss function of a VAD model, the model is inclined toward achieving more robust anomaly detection.

Temporal smoothing constraint

The temporal smoothing constraint is a crucial component of loss functions in VAD, employed to maintain stability in predicted anomaly scores across consecutive frames. This constraint aims to mitigate abrupt fluctuations in anomaly scores between adjacent frames, aligning with the expectation that successive frames typically exhibit similar anomaly characteristics. The temporal smoothing constraint penalizes sudden changes in the anomaly score between sequential frames as follows in Eq. 11:

$$L_{smooth} = \lambda_1 \sum_{i=1}^{|\mathcal{B}_+|} \left(S(V_i;\theta) - S(V_{i+1};\theta) \right)^2 \tag{11}$$

where a λ_1 is a smoothing coefficient, $S(V_i;\theta)$ is the anomaly score of frame i, and $S(V_{i+1};\theta)$ is the anomaly score of frame $i+1$.

Sparsity constraint

The sparsity constraint is a commonly used regularization term in VAD that forces the anomalous frames to be a minority among the video frames. In VAD, the anomalous frames are generally assumed to be the minority compared to the normal frames. The sparsity constraint penalizes the total number of anomalous frames in the video as follows in Eq. 12:

$$L_{sparsity} = \lambda_2 \sum_{V \in \mathcal{B}_+} S(V_i;\theta) \tag{12}$$

where a λ_2 is a sparsity coefficient and $S(V_i;\theta)$ is the anomaly score of frame i.

While spatiotemporal constraints are well-suited for handling the abrupt anomalies within the video sequence, the recent trends toward incorporating VAD-specific constraints improve the overall performance. Future directions in this domain include the following:

- Adversarial regularization techniques may help the model learn more resilient features by introducing adversarial perturbations during training, thereby enhancing its ability to detect anomalies in the presence of adversarial manipulations.
- Incorporating temporal regularization techniques to enforce temporal consistency and coherence in anomaly detection predictions over time can encourage smooth transitions and consistent predictions across consecutive frames.
- Graph-based regularization techniques can impose structural constraints on the learned representations, encouraging the model to capture meaningful interactions and contextual information among objects, scenes, or events, leading to more accurate anomaly detection.
- Integrating sparse regularization techniques to encourage sparsity in the learned representations, focusing the model's attention on salient features or regions within videos can promote the selection of informative features while suppressing irrelevant noise.

- Addressing catastrophic forgetting and model degradation over time by incorporating continual learning regularization techniques can enable the model to adapt and learn from new data while preserving previously learned knowledge, facilitating the adaptation in surveillance environments.

5.7 Anomaly score

The anomaly score indicates the likelihood of a segment or frame in a video being abnormal, calculated based on how much it deviates from normal patterns. A high anomaly score suggests a high probability of an anomaly.

For the reconstruction approaches [20, 79], after training the model, its effectiveness is measured by inputting test data to see if it can accurately identify unusual activities with a minimal number of false alarm rates. Then, calculate the anomaly score $S(V_i)$ for each frame V_i by scaling between 0 and 1 using the reconstruction error of the frame $e(V_i)$.

$$S(V_i) = \frac{e(V_i) - \min_{V_i} e(V_i)}{\max_{V_i} e(V_i)}. \tag{13}$$

Hence, we can define the regularity score to be:

$$S_r(V_i) = 1 - S(V_i). \tag{14}$$

However, for future frame prediction methods, [104] has shown that peak signal-to-noise ratio (PSNR) is a superior method for evaluating image quality [52, 55], as indicated below:

$$\mathrm{PSNR}(Y, \hat{Y}) = 10 \log_{10} \left(\frac{\max_{\hat{Y}}^2}{\frac{1}{N} \sum_{i=0}^{N} (Y_i - \hat{Y}_i)^2} \right), \tag{15}$$

The assumption is that we can make accurate predictions for normal events. Therefore, we can detect anomalies by measuring the difference between a predicted frame $\hat{Y}$ and its corresponding ground truth Y. Where $\max_{\hat{Y}}$ is the maximum possible value of the image intensities divided by the mean square error (MSE). In image reconstruction tasks, a higher PSNR would indicate a lower error and, thus, higher fidelity to the original image, which is the desired outcome.

After calculating the PSNR between the predicted $\hat{Y}$ and its corresponding ground truth Y for each frame V_i in a test video and normalizing these values, each frame's regularity score is determined $S_r(V_i)$ using the equation below.

$$S_r(V_i) = \frac{\mathrm{PSNR}(Y_{V_i}, \hat{Y}_{V_i}) - \min_{V_i} \mathrm{PSNR}(Y_{V_i}, \hat{Y}_{V_i})}{\max_{V_i} \mathrm{PSNR}(Y_{V_i}, \hat{Y}_{V_i}) - \min_{V_i} \mathrm{PSNR}(Y_{V_i}, \hat{Y}_{V_i})} \tag{16}$$

This score indicates the likelihood of a frame being normal or abnormal, with the potential to set a threshold point for distinguishing between the two.

6 Model evaluation

6.1 Datasets guideline

As discussed in Sect. 5.1, several datasets were used in the literature for training and evaluating VAD models. These datasets are well-suited for different types of supervision schemes. Most of the datasets, including UCF-Crime [15], ShanghaiTech [18], XD-Violence [17], and CUHK Avenue [19], provide video-level labels for the training set, which could be utilized for training weakly supervised VAD as in the works by Joo et al. [30] and Cho et al. [103]. These datasets could also be utilized for semi-supervised VAD by utilizing the training set without the video-level labels as in the works of Shi at al. [105] and Zaheer et al. [81] and for self-supervised learning as in the work of Zhou et al. [77].

6.2 Evaluation metrics

Most previous works mainly compared their results on different datasets using metrics such as the area under the ROC curve (AUC) of the frame level, equal error rate (EER), and average precision (AP) of the frame level to measure performance. The main aim of these metrics is to assess the model's proficiency in differentiating between normal and abnormal videos.

Area Under the ROC Curve (AUC):

The AUC score is an important metric for evaluating VAD models' performance, which is evaluated by calculating the area under the receiver operating characteristic (ROC) curve. The ROC curve plots the true positive rate (TPR) against the false positive rate (FPR) while varying the threshold value. The TPR and FPR are found using the following equations:

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{17}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{FP} + \mathrm{TN}} \tag{18}$$

Achieving AUC scores closer to one indicates better anomaly detection capability, while a random classifier model will achieve an AUC score of 0.5. Two types of the AUC score are Micro-AUC and Macro-AUC, in which the first considers all class samples together for a single score while the second evaluates each class individually and averages the scores, respectively. The Micro-AUC is generally more affected by the majority class, while the Macro-AUC provides a fairer evaluation across all classes.

Average Precision (AP):

The AP score evaluates the precision value at different recall values and calculates the average. It summarizes the precision-recall curve into a single value representing average performance across all threshold levels.

Equal Error Rate (EER): The EER represents the point at which the FPR equals the false negative rate (FNR). Lower EER values indicate a greater degree of accuracy in the system [77].

In this survey, we use the AUC as the performance metric in our quantitative comparison of the SOTA models in Sect. 7.1.

7 Comparative analysis of SOTA models

7.1 Quantitative comparison

There have been several works in the literature that proposed different architectures of deep learning models for addressing the problem of VAD. Table 3 highlights a benchmarking of recent anomaly detection works on

Table 3 Performance benchmarking of recent anomaly detection methods on publicly available datasets. Reported values represent the area under the curve (AUC, %) for each dataset

Sup. type	Method	Venue	Year	U-C	STech	XD	CUHK
Self	SS-MTL [13]	CVPR	2021	–	–	–	86.9
	EVAL [48]	CVPR	2023	–	76.63	–	86.02
	Zhang et al. [106]	CVPR	2024	**80.60**	**87.50**	–	**94.30**
Unsup	GCL [81]	CVPR	2022	71.04	**78.93**	–	–
	CLAP [107]	CVPR	2024	**85.50**	–	**90.04**	–
Semi	Lu et al. [19]	ICCV	2013	65.51	68.00	–	–
	Hasan et al. [20]	CVPR	2016	50.60	60.85	–	70.20
	AMC [72]	ICCV	2019	–	–	–	86.9
	MemAE [108]	ICCV	2019	–	71.2	–	83.3
	MPED-RNN [109]	CVPR	2019	–	73.4	–	–
	CVAE [24]	ICCV	2021	–	76.2	–	91.1
	USTN-DSC [60]	CVPR	2023	–	73.8	–	89.9
	FPDM [110]	ICCV	2023	**74.70**	78.60	–	–
	SLMPT [105]	ICCV	2023	–	78.8	–	90.9
	MGAN-CL [25]	TCSVT	2024	–	73.6	–	–
	Ristea et al. [111]	CVPR	2024	–	79.10	–	**91.30**
	SSAE [112]	TPAMI	2024	–	**80.50**	–	90.20
Weakly	Sultani et al. [15]	CVPR	2018	75.41	–	–	–
	TCN-IBL [32]	ICIP	2019	78.66	82.50	–	–
	GCN-TSN [33]	CVPR	2019	82.12	84.44	–	–
	SRF [34]	IEEE SPL	2020	79.54	84.16	–	–
	Noise cleaner [84]	CVPRW	2021	78.26	84.16	–	–
	DAM [35]	AVSS	2021	82.67	88.22	–	–
	SA-CRF [36]	ICCV	2021	85.00	96.85	–	–
	RTFM [37]	ICCV	2021	84.30	97.21	77.81	–
	MIST [53]	CVPR	2021	82.30	94.84	–	–
	MSLNet-CTE [27]	AAAI	2022	85.62	97.32	78.59	–
	CUPL [97]	CVPR	2023	86.22	–	–	–
	HSC [26]	CVPR	2023	–	83.4	–	**93.7**
	UMIL [96]	CVPR	2023	86.75	–	–	–
	TeD-SPAD [113]	ICCV	2023	75.06	–	–	–
	CLAV [103]	CVPR	2023	86.10	97.60	81.30	89.80
	TEVAD [31]	CVPRW	2023	84.90	98.10	79.80	–
	CLIP-TSA [30]	ICIP	2023	87.58	98.32	82.19	–
	TPWNG [74]	CVPR	2024	87.79	–	**83.68**	–
	Chen et al. [114]	CVPR	2024	86.83	98.35	–	–
	FDPN [115]	WACV	2025	**88.03**	**98.51**	–	–

Bold values indicate the highest performance for each dataset within the corresponding supervision type

U-C.—UCF-Crime, STech—ShanghaiTech, XD.—XD-Violence, CUHK.—CUHK Avenue, Sup. Type—Supervision Type, Self.—Self-Supervised, Unsup.—Unsupervised, Semi.—Semi-Supervised, Weakly—Weakly Supervised

publicly available datasets. The table offers a detailed look at the evolution and current state of anomaly detection techniques, primarily focusing on publicly available datasets including UCF-Crime, ShanghaiTech, XD-Violence, and Avenue. The data span a decade, from 2013 to 2023, providing a comprehensive view of the field's progression in the last ten years. An important trend that can be observed in the table is the shift in research interest from supervised VAD approaches to weakly supervised VAD learning methods. This change, particularly notable from 2018 onward and greatly influenced by the work of Sultani et al. [15] who introduced the "UCF-Crime" dataset and the multi-instance learning (MIL) learning pipeline, suggests a growing preference in the industry for techniques that require less fully labeled data, which is often expensive or difficult to obtain, compared to weakly supervised approaches that rely on video-level labels. This shift reflects the practical challenges and evolving needs in anomaly detection applications, particularly in applications where hundreds of hours of unlabeled or weakly labeled video footage is available.

One important consideration is the variety of feature extractors that were used in the literature. These methods included simpler methods such as principal component analysis (PCA) as well as more complex methods such as 3D convolutional networks (C3D) and inflated 3D ConvNet (I3D). This highlights the difficulty of feature extraction for achieving robust representative characteristics of anomalous patterns in videos.

The reported AUC scores across different datasets highlight the prominent enhancements to VAD systems over time, indicating the effectiveness of recent VAD systems in real-world conditions. Recent models such as FDPN [115] achieved exceptionally high AUC scores.

A novel approach that utilized the vision-language models is TPWNG [74], published in 2024, which outperformed the CLIP-TSA over two datasets as shown in Table 3. Based on the SOTA comparison, it is evident that vision-language-based approaches are currently at the forefront of the VAD research.

However, the analysis reveals several instances of missing data. These gaps across various methodologies and datasets highlight the necessity for more comprehensive benchmarking efforts. Addressing such gaps is crucial to foster a deeper understanding and facilitate meaningful comparisons among different anomaly detection methods, ultimately advancing the field.

7.2 Qualitative comparison

To observe in more detail the performance of current state-of-the-art models on various normal and anomalous surveillance videos, this section presents a qualitative analysis of these models' performance. Figure 6 depicts a qualitative assessment of correctly classified and misclassified video frames using four VAD models, namely Sultani et al. [15], GCN [81], CLAV [103], and VAD-CLIP [39]. As shown in the figure's first row, these four VAD models were able to correctly classify normal and anomalous video frames taken from different scenes and environments. However, these models misclassified other similar frames taken from the same videos, either by detecting anomalies in a normal video or by not detecting an anomalous video segment. For example, the model proposed by Sultani et al. [15] misclassified the frame number 10000 in the video "Burglary032," which shows a person entering an office through a window, by labeling it as "Normal" while it is an "Anomaly." This was attributed to the immense darkness of the scene, which made recognizing such an ambiguous anomaly difficult. Another example of misclassification is with the GCN model by Zaheer et al. [81] as it misclassified the frame number 4400 in the video "Stealing058," which shows a car leaving the parking spot, by labeling it as an "Anomaly" while it is "Normal." This could be attributed to the sudden movement of the car with its lights turned on. Similarly, the model proposed by Cho et al. [103] named CLAV misclassified the initial frames of a scene showing cars stopping and accelerating again in the street. This increased anomaly score could be explained by the abrupt stopping of the cars, which in many cases could be considered an anomaly. Finally, the VAD-CLIP model proposed by Wu et al. [39] misclassified the frame number 360 in the video "Shooting008," which shows a person crawling on the floor, by classifying it as an "Anomaly" while it is "Normal." This misclassification could be explained as the person crawling moves unconventionally in contrast with what commonly happens in normal street and shopping center scenes. It could be concluded that some normal and abnormal frames could be

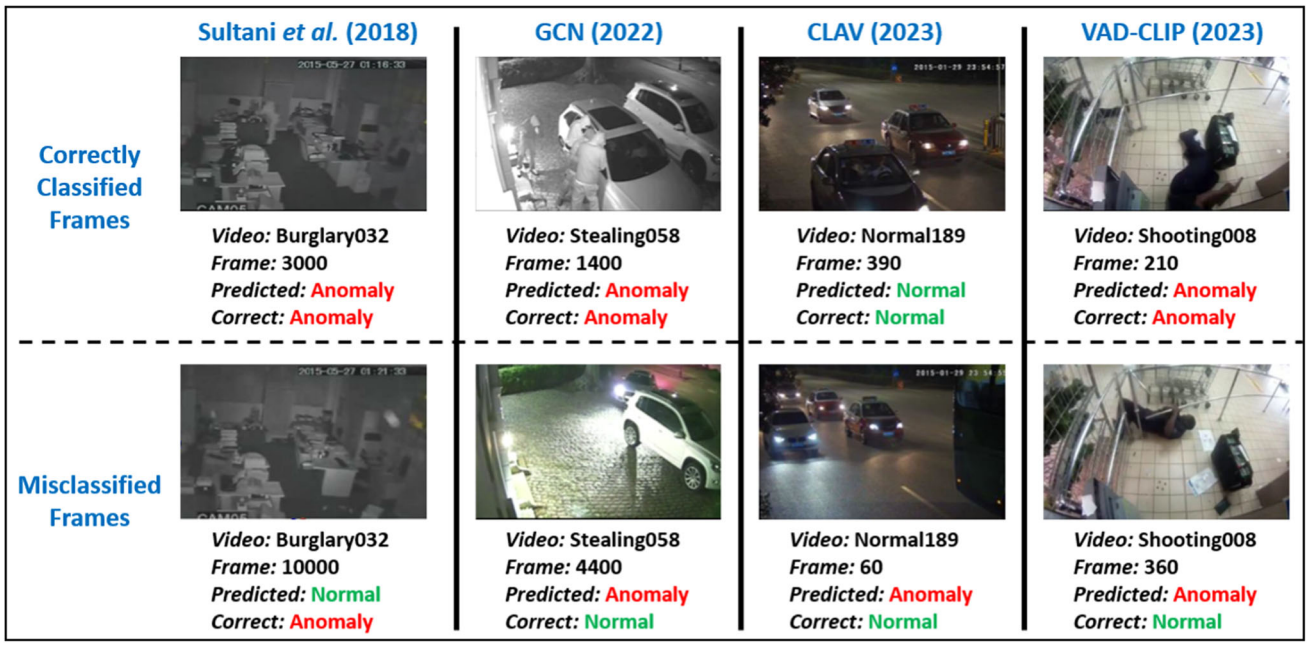

Fig. 6 A qualitative comparison and illustration of correctly and incorrectly classified frames using four VAD models, namely Sultani et al. [15], GCN [81], CLAV [103], and VAD-CLIP [39]

misclassified when the overall context of the video cannot be inferred directly from the frame, in cases where the frames give a false impression of a different context.

7.2.1 Model interpretability and explainability

In VAD, model interpretability and explainability are key concerns, especially for real-world applications like surveillance, where understanding the reasons behind a system's decision is crucial. A model not only detects anomalies but also provides human-understandable reasons for why certain events are flagged as abnormal; it improves decision-making and allows for easier validation of the system's behavior.

Hinami et al. propose [116] a joint detection and recounting framework that emphasizes the recounting of abnormal events. Their method enhances explainability by learning deep, generic knowledge about objects, actions, and attributes through CNNs, which can be mapped directly to human-understandable concepts. For instance, when an anomaly is detected, the system explains it by identifying relevant objects (e.g., "person"), actions (e.g., "running"), and attributes (e.g., "young"), thereby offering a clear rationale for the anomaly. This recounting aspect is critical because it not only flags an event as abnormal but also provides context for why it was considered unusual. However, this method's reliance on object proposals can result in missed detections when certain objects or actions are not represented in the training data, limiting the comprehensiveness of the recounting in complex scenarios.

Another approach of Singh *et al.* [48] is based on explainability by introducing the EVAL framework, which focuses on high-level appearance and motion features for the localization of video anomalies. This system is explicitly designed to generate human-interpretable explanations for its decisions. By training deep networks to extract features related to object types, motion directions, and speeds, EVAL provides spatial and temporal justifications for detected anomalies. For instance, the system can explain an anomaly by stating that a car was detected on the sidewalk instead of the road, or that an object moved in an unexpected direction. The significant advantage of this method lies in its ability to provide specific, scene-dependent explanations without retraining the model for each new environment, making it a flexible and efficient solution. The explainability here is

grounded in comparing the detected event with a predefined set of normal behaviors, which ensures that the anomaly detection process is interpretable to human users.

Wu et al. [117] took a different approach by incorporating SHapley Additive exPlanations (SHAP) [118] with a deep learning-based video anomaly detection framework. Their method integrates SHAP with a denoising autoencoder, making it one of the few models that directly focuses on post hoc interpretability in video anomaly detection. SHAP provides a detailed breakdown of the contributions of different features (such as object classifications, motion patterns, or background segmentation) to the model's predictions. This level of explanation allows operators to understand why specific frames or events were classified as anomalous. For example, SHAP might indicate that an event was considered abnormal because of a sudden change in the object's motion vector, making the decision-making process more transparent. This approach is particularly valuable because it shifts from purely model-driven explanations to user-centric interpretations, thus reducing the "black box" nature of many deep learning models. In comparison, Hinami et al.'s model emphasizes real-time recounting based on learned visual concepts, while Singh et al.'s method provides a spatial and contextual explanation through high-level scene modeling. In contrast, Wu et al. utilize SHAP to offer feature-based interpretability, focusing on post hoc explanations that attribute the model's decision to specific features within the video. These methodologies provide valuable pathways for future work in making anomaly detection not only accurate but also deeply interpretable for end-users.

Additionally, vision-language model-based approaches emerge as valuable resources for enhancing model interpretability and explainability in video anomaly detection. By focusing on assigning semantic meanings to video content in a human-understandable manner—much like captioning models—these approaches enable systems to provide clear, language-based explanations for detected anomalies. An example of that is when a VLM could generate an explanation of the anomaly sequence, such as "A car is driving on the pedestrian path." This explanatory text clarifies why the model considers that specific instance as anomalous while also making it easier for human operators to understand the justification of both correct and incorrect predictions. By translating complex visual data into descriptive language, VLM-based approaches bridge the gap between machine perception and human interpretation, thereby enhancing the transparency and trustworthiness of the anomaly detection system. Compared to methods that rely solely on feature attribution or object detection, VLMs provide more comprehensive and context-aware explanations, leveraging rich semantic understanding to facilitate better decision-making in real-world applications. As in the recent work by Yang et al., they proposed the first rule-based reasoning for one-class VAD.

8 Visualizing bibliometric networks for thematic analysis

In bibliometrics, visualization emerges as a potent technique for analyzing diverse networks, encompassing citation-based, co-authorship-based, or keyword co-occurrence-based networks. Density visualizations provide a rapid overview of the essential components within a bibliometric network. Employing NLP techniques, one can construct term co-occurrence networks for textual data in English. The rationale behind employing interconnected networks is to illustrate the temporal and relevance-driven evolution of the field.

We utilized VOSviewer [119], a tool employing a distance-based method to generate visualizations of bibliometric networks. Directed networks, like those formed by citation relationships, are treated as undirected in this context. VOSviewer autonomously organizes network nodes into clusters, where nodes with similar attributes are grouped. Each node in the network is neatly assigned to one of these clusters. Furthermore, VOSviewer employs color to denote a node's membership within a cluster in bibliometric network visualizations.

Five distinct clusters have emerged, as shown in Fig. 7, each representing a different research theme. The green cluster, centered around the theme of diffusion, explores methods based on diffusion models for anomaly detection, with representative work such as "Anomaly Detection in Satellite Videos using Diffusion Models" [120]. In the yellow cluster, efficiency takes precedence, with research efforts directed toward developing

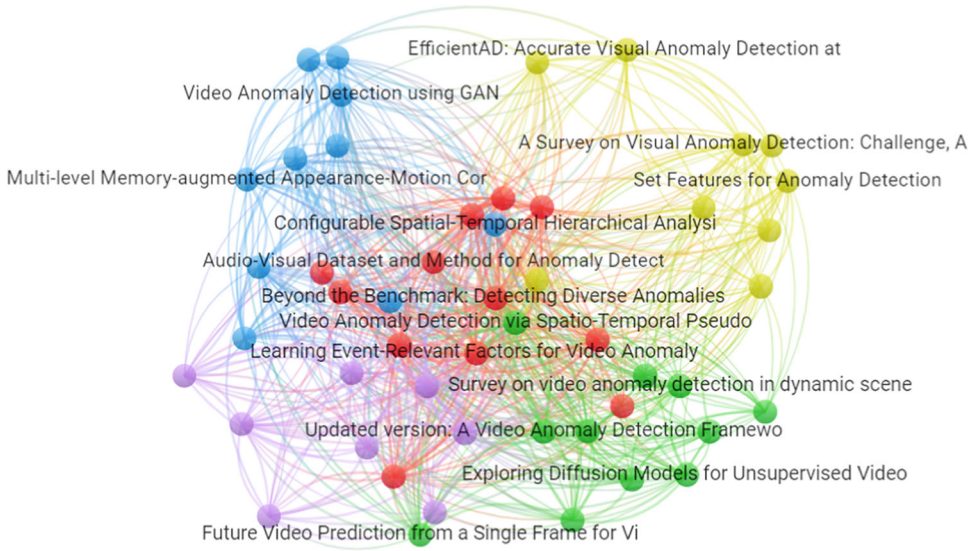

Fig. 7 Visualizing bibliometric networks for thematic analysis of recent literature (50 top-cited papers) on video anomaly detection between the years 2023–2024

algorithms capable of accurate anomaly detection at millisecond-level latencies, as demonstrated in the paper "EfficientAD: Accurate Visual Anomaly Detection at Millisecond-Level Latencies" [121]. The black cluster focuses on interleaving one-class and weakly supervised models with adaptive thresholding for unsupervised video anomaly detection, exemplified by the paper "Interleaving One-Class and Weakly-Supervised Models with Adaptive Thresholding for Unsupervised Video Anomaly Detection" [122]. Within the red cluster, the exploration delves into vision-language models (VLM), particularly CLIP, for video anomaly recognition, as seen in "Delving into CLIP latent space for Video Anomaly Recognition" [123]. Finally, the black cluster is dedicated to research involving generative adversarial networks (GANs) for VAD tasks, with representative work titled "Video Anomaly Detection using GAN" [124]. All clusters are further illustrated in Table 4.

9 Discussion and future research

9.1 Solved and unsolved challenges

In this survey paper, we present a comprehensive review that serves as a guideline to solve challenges related to VAD. The past decade has witnessed a remarkable evolution in the field of video anomaly detection (VAD), marked by a notable transition from supervised to weakly supervised learning approaches and reconstruction-based techniques. This shift reflects a growing preference for methods capable of operating effectively with less reliance on fully labeled data, which can be both costly and challenging to obtain. Notably, the majority of

Table 4 Cluster central point and color for each cluster with representative paper and theme

Cluster	Research paper: the title
●	Anomaly Detection in Satellite Videos using Diffusion Models (Theme: Diffusion)
●	Delving into CLIP latent space for Video Anomaly Recognition (Theme: VLM)
●	EfficientAD: Accurate Visual Anomaly Detection at Millisecond-Level Latencies (Theme: Efficiency)
●	Video Anomaly Detection using GAN (Theme: GAN)
●	Interleaving One-Class and Weakly-Supervised Models with Adaptive Thresholding for Unsupervised Video Anomaly Detection (Theme: Weakly-Supervised)

benchmarking datasets are weakly supervised, featuring video-level annotations, while reconstruction approaches utilize only normal data in an unsupervised manner during training.

The existing literature presents a diverse array of VAD datasets, ranging from specific single-scene datasets like UCSD Pedestrian to more comprehensive, multi-scene collections such as UCF-Crime and XD-Violence. These benchmark datasets exhibit notable differences regarding their total durations, number of videos, and types of anomalies covered.

Regardless of the fact that tremendous effort was made in recent years on developing datasets for VAD, some crucial challenges remain. These challenges include the limited diversity in the covered environments in videos, the restricted range of anomalies covered, and the evident class imbalance. Such challenges noticeably affect the generalization ability and detection accuracy of trained VAD models under different supervision settings. Additionally, the use of a mixed-method approach for extracting features for VAD is an important direction for improvement, which illustrates the complexity of VAD as a computer vision task. Such techniques that could be hybridized include CNNs, AEs, GANs, and VLMs. This necessity emphasizes the difficulty in extracting meaningful features for VAD, as each model has its strengths and weaknesses.

Furthermore, the utilization of appropriate loss functions is crucial for building effective VAD systems as it directly influences the model's prediction accuracy. In addition, the use of appropriate regularization terms in the loss function further enhances the model's capabilities by improving its generalizability after training. Examples of such important regularization terms are the sparsity and smoothness constraints. After training VAD models, evaluation of anomaly score predictions has been carried out using metrics such as AUC, AP, and EER, which gives insights into the quality of predictions and thus the performance limitations. It is necessary to evaluate VAD models on multiple datasets with different settings to assess their reliability and adaptability.

Another crucial challenge is the difficulty of the practical implementation of deep learning-based VAD systems. Even though these models were able to achieve outstanding performance on VAD benchmarks, most of these approaches do not take the computational cost into consideration despite the high dimensionality of video data. Most recent models utilize extremely sophisticated deep learning architecture, which results in an apparent trade-off between accuracy and inference speed. Developing lightweight VAD systems capable of generating accurate predictions in real-time will drive this technology closer to practical implementation and commercialization.

9.2 Prospective methods

Future works in the field of VAD could considerably benefit from recent works in the fields of *Graph Signal Processing (GSP)* and *Graph Spectral Clustering*. Such types of graph-based methods have the ability to model highly complex dependencies between extracted features from both consecutive and separated frames of video, which helps in identifying anomalous patterns. These methods treat each video frame as a graph's node and analyze the feature dependencies over time to capture anomalies in the spatiotemporal domain.

Among these methods, graph spectral clustering allows for analyzing and segmenting video data of extremely high dimensionality, where complex scenarios are clustered based on the representative spectral properties. As this method links the features extracted from several data points at a time, it could be of crucial benefit to boosting VAD performance, particularly in cases of difficult-to-detect anomalies.

Moreover, the hybridization of state-of-the-art VAD methods with the recent advancements in vision-language models (VLMs) is a prominent direction for future investigation. The inclusion of vision-language features could be of tremendous benefit as they can provide more meaningful features by linking the visual content of the video frame with the real-world semantic and lingual understanding of LLMs.

Despite the huge progress that was made on the VAD problem in the last years, datasets with wider diversity in content and larger data size are critically required. Specifically, datasets covering a broader range of scenarios, encompassing multiple scenes and anomalies, are essential. Such datasets would not only facilitate the

benchmarking of existing models but also stimulate innovation by presenting researchers with more challenging real-world situations to address.

Another promising direction for VAD is the incorporation of comprehensive textual descriptions as an easy-to-interpret form of extracted features. Video description and captioning models have shown outstanding improvement in the last years, being able to accurately and extensively describe the contents of images and videos in a human-understandable language [125]. The use of such features could significantly boost the explainability of such anomaly detection and localization models and convert the deep learning-based VAD from a "black box" of incomprehensible features to a more interpretable form.

Furthermore, since VLMs are emerging, an important direction for future research involves the integration of textual descriptions in the data containing contextual details, such as frame-level captions, into anomaly detection models. This integration has the potential to significantly improve model performance by providing rich, descriptive information that aids in the interpretation and analysis of visual content. Consequently, this trend will necessitate the development of stronger loss functions capable of more effectively managing textual information.

In conclusion, this survey paper highlights an exciting phase of growth and transformation in the field of VAD, characterized by methodological advancements, the integration of new technologies, and a shift toward more efficient learning approaches. However, the identified gaps and challenges underscore the need for continued efforts within the research community to develop comprehensive datasets and explore novel methodologies, ultimately advancing the state-of-the-art in anomaly detection.

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Data availability This is a review article and does not involve the generation or analysis of new datasets. All data referenced in this work are available in the cited publications.

Declarations

Conflict of interest There are no conflict of interest from the authors.

Ethical approval There is no any ethical conflict.

References

1. Ramachandra B, Jones MJ, Vatsavai RR (2020) A survey of single-scene video anomaly detection. IEEE Trans Pattern Anal Mach Intell 44(5):2293–2312
2. Aggarwal CC, Aggarwal CC (2013) Applications of outlier analysis. Outlier Anal 373–400
3. Fang X, Guo W, Li Q, Zhu J, Chen Z, Yu J, Zhou B, Yang H (2020) Sewer pipeline fault identification using anomaly detection algorithms on video sequences. IEEE Access 8:39574–39586
4. Ukil A, Bandyoapdhyay S, Puri C, Pal A (2016) Iot healthcare analytics: the importance of anomaly detection. In: 2016 IEEE 30th International conference on advanced information networking and applications (AINA). IEEE, pp 994–997
5. Dereszynski EW, Dietterich TG (2011) Spatiotemporal models for data-anomaly detection in dynamic environmental monitoring campaigns. ACM Trans Sensor Netw (TOSN) 8(1):1–36
6. Baccari S, Hadded M, Ghazzai H, Touati H, Elhadef M (2024) Anomaly detection in connected and autonomous vehicles: a survey, analysis, and research challenges. IEEE Access
7. Jiang M, Hou C, Zheng A, Hu X, Han S, Huang H, He X, Yu PS, Zhao Y (2023) Weakly supervised anomaly detection: a survey. arXiv preprint arXiv:2302.04549
8. Pang G, Shen C, Cao L, Hengel AVD (2021) Deep learning for anomaly detection: a review. ACM Comput Surv (CSUR) 54(2):1–38
9. Shao Z, Bian H, Chen Y, Wang Y, Zhang J, Ji X et al (2021) Transmil: transformer based correlated multiple instance learning for whole slide image classification. Adv Neural Inf Process Syst 34:2136–2147
10. Ullah W, Hussain T, Ullah FUM, Lee MY, Baik SW (2023) Transcnn: hybrid CNN and transformer mechanism for surveillance anomaly detection. Eng Appl Artif Intell 123:106173

11. Nayak R, Pati UC, Das SK (2021) A comprehensive review on deep learning-based methods for video anomaly detection. Image Vis Comput 106:104078
12. Kiran BR, Thomas DM, Parakkal R (2018) An overview of deep learning based methods for unsupervised and semi-supervised anomaly detection in videos. J Imaging 4(2):36
13. Georgescu M-I, Barbalau A, Ionescu RT, Khan FS, Popescu M, Shah M (2021) Anomaly detection in video via self-supervised and multi-task learning. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 12742–12752
14. Liu G, Shu L, Yang Y, Jin C (2023) Unsupervised video anomaly detection in UAVs: a new approach based on learning and inference. Front Sustain Cities 5:1197434
15. Sultani W, Chen C, Shah M (2018) Real-world anomaly detection in surveillance videos. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 6479–6488
16. Nawaratne R, Alahakoon D, De Silva D, Yu X (2019) Spatiotemporal anomaly detection using deep learning for real-time video surveillance. IEEE Trans Industr Inf 16(1):393–402
17. Wu P, Liu J, Shi Y, Sun Y, Shao F, Wu Z, Yang Z (2020) Not only look, but also listen: learning multimodal violence detection under weak supervision. In: Computer vision—ECCV 2020: 16th European conference, glasgow, UK, Aug 23–28, 2020, Proceedings, Part XXX 16. Springer, pp 322–339
18. Luo W, Liu W, Gao S (2017) A revisit of sparse coding based anomaly detection in stacked RNN framework. In: 2017 IEEE international conference on computer vision (ICCV), pp 341–349. https://doi.org/10.1109/ICCV.2017.45
19. Lu C, Shi J, Jia J (2013) Abnormal event detection at 150 fps in matlab. In: Proceedings of the IEEE international conference on computer vision, pp 2720–2727
20. Hasan M, Choi J, Neumann J, Roy-Chowdhury AK, Davis LS (2016) Learning temporal regularity in video sequences. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 733–742
21. Dalal N, Triggs B (2005) Histograms of oriented gradients for human detection. In: 2005 IEEE computer society conference on computer vision and pattern recognition (CVPR'05), vol 1. IEEE, pp 886–893
22. Dalal N, Triggs B, Schmid C (2006) Human detection using oriented histograms of flow and appearance. In: Computer vision—ECCV 2006: 9th European conference on computer vision, Graz, Austria, May 7–13, 2006. Proceedings, Part II 9. Springer, pp 428–441
23. Luo W, Liu W, Gao S (2017) Remembering history with convolutional LSTM for anomaly detection. In: 2017 IEEE international conference on multimedia and expo (ICME). IEEE, pp 439–444
24. Liu Z, Nie Y, Long C, Zhang Q, Li G (2021) A hybrid video anomaly detection framework via memory-augmented flow reconstruction and flow-guided frame prediction. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 13588–13597
25. Li D, Nie X, Gong R, Lin X, Yu H (2023) Multi-branch Gan-based abnormal events detection via context learning in surveillance videos. IEEE Trans Circuits Syst Video Technol
26. Sun S, Gong X (2023) Hierarchical semantic contrast for scene-aware video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 22846–22856
27. Li S, Liu F, Jiao L (2022) Self-training multi-sequence learning with transformer for weakly supervised video anomaly detection. Proc AAAI Conf Artif Intell 36:1395–1403
28. Ni B, Peng H, Chen M, Zhang S, Meng G, Fu J, Xiang S, Ling H (2022) Expanding language-image pretrained models for general video recognition. In: European conference on computer vision. Springer, pp 1–18
29. Ju C, Han T, Zheng K, Zhang Y, Xie W (2022) Prompting visual-language models for efficient video understanding. In: European conference on computer vision. Springer, pp 105–124
30. Joo HK, Vo K, Yamazaki K, Le N (2023) Clip-tsa: clip-assisted temporal self-attention for weakly-supervised video anomaly detection. In: 2023 IEEE international conference on image processing (ICIP). IEEE, pp 3230–3234
31. Chen W, Ma KT, Yew ZJ, Hur M, Khoo DA-A (2023) Tevad: improved video anomaly detection with captions. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 5548–5558
32. Zhang J, Qing L, Miao J (2019) Temporal convolutional network with complementary inner bag loss for weakly supervised anomaly detection. In: 2019 IEEE international conference on image processing (ICIP). IEEE, pp 4030–4034
33. Zhong J-X, Li N, Kong W, Liu S, Li TH, Li G (2019) Graph convolutional label noise cleaner: Train a plug-and-play action classifier for anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 1237–1246
34. Zaheer MZ, Mahmood A, Shin H, Lee S-I (2020) A self-reasoning framework for anomaly detection using video-level labels. IEEE Signal Process Lett 27:1705–1709
35. Majhi S, Das S, Brémond F (2021) Dam: dissimilarity attention module for weakly-supervised video anomaly detection. In: 2021 17th IEEE international conference on advanced video and signal based surveillance (AVSS). IEEE, pp 1–8
36. Purwanto D, Chen Y-T, Fang W-H (2021) Dance with self-attention: a new look of conditional random fields on anomaly detection in videos. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 173–183

37. Tian Y, Pang G, Chen Y, Singh R, Verjans JW, Carneiro G (2021) Weakly-supervised video anomaly detection with robust temporal feature magnitude learning. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 4975–4986
38. Radford A, Kim JW, Hallacy C, Ramesh A, Goh G, Agarwal S, Sastry G, Askell A, Mishkin P, Clark J et al (2021) Learning transferable visual models from natural language supervision. In: International conference on machine learning. PMLR, pp 8748–8763
39. Wu P, Zhou X, Pang G, Zhou L, Yan Q, Wang P, Zhang Y (2024) Vadclip: adapting vision-language models for weakly supervised video anomaly detection. Proc AAAI Conf Artif Intell 38:6074–6082
40. Lin K, Li L, Lin C-C, Ahmed F, Gan Z, Liu Z, Lu Y, Wang L (2022) Swinbert: end-to-end transformers with sparse attention for video captioning. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 17949–17958
41. Chalapathy R, Chawla S (2019) Deep learning for anomaly detection: a survey. arXiv preprint arXiv:1901.03407
42. Baradaran M, Bergevin R (2024) A critical study on the recent deep learning based semi-supervised video anomaly detection methods. Multimedia Tools Appl 83(9):27761–27807
43. Choudhry N, Abawajy J, Huda S, Rao I (2023) A comprehensive survey of machine learning methods for surveillance videos anomaly detection. IEEE Access
44. Liu Y, Yang D, Wang Y, Liu J, Liu J, Boukerche A, Sun P, Song L (2023) Generalized video anomaly event detection: systematic taxonomy and comparison of deep models. ACM Comput Surveys
45. Adam A, Rivlin E, Shimshoni I, Reinitz D (2008) Robust real-time unusual event detection using multiple fixed-location monitors. IEEE Trans Pattern Anal Mach Intell 30(3):555–560. https://doi.org/10.1109/TPAMI.2007.70825
46. Mahadevan V, Li W, Bhalodia V, Vasconcelos N (2010) Anomaly detection in crowded scenes. In: 2010 IEEE computer society conference on computer vision and pattern recognition, pp 1975–1981. https://doi.org/10.1109/CVPR.2010.5539872
47. Ramachandra B, Jones M (2020) Street scene: a new dataset and evaluation protocol for video anomaly detection. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp 2569–2578
48. Singh A, Jones MJ, Learned-Miller EG (2023) Eval: explainable video anomaly localization. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 18717–18726
49. Cao C, Lu Y, Wang P, Zhang Y (2023) A new comprehensive benchmark for semi-supervised video anomaly detection and anticipation. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR), pp 20392–20401
50. Sabzalian B, Marvi H, Ahmadyfard A (2019) Deep and sparse features for anomaly detection and localization in video. In: 2019 4th International conference on pattern recognition and image analysis (IPRIA). IEEE, pp 173–178
51. Cheng K-W, Chen Y-T, Fang W-H (2015) Video anomaly detection and localization using hierarchical feature representation and gaussian process regression. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 2909–2917
52. Liu W, Luo W, Lian D, Gao S (2018) Future frame prediction for anomaly detection—a new baseline. In: Proceedings of the IEEE conference on computer vision and pattern recognition (CVPR), pp 6536–6545
53. Feng J-C, Hong F-T, Zheng W-S (2021) Mist: multiple instance self-training framework for video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 14009–14018
54. Chen H, Mei X, Ma Z, Wu X, Wei Y (2023) Spatial-temporal graph attention network for video anomaly detection. Image Vis Comput 131:104629
55. Luo W, Liu W, Lian D, Gao S (2021) Future frame prediction network for video anomaly detection. IEEE Trans Pattern Anal Mach Intell 44(11):7505–7520
56. Zhong Y, Chen X, Hu Y, Tang P, Ren F (2022) Bidirectional spatio-temporal feature learning with multiscale evaluation for video anomaly detection. IEEE Trans Circuits Syst Video Technol 32(12):8285–8296
57. Mansour RF, Escorcia-Gutierrez J, Gamarra M, Villanueva JA, Leal N (2021) Intelligent video anomaly detection and classification using faster RCNN with deep reinforcement learning model. Image Vis Comput 112:104229
58. Chen Z, Duan J, Kang L, Qiu G (2022) Supervised anomaly detection via conditional generative adversarial network and ensemble active learning. IEEE Trans Pattern Anal Mach Intell 45(6):7781–7798
59. Abdalla M, Hassan H, Mostafa N, Abdelghafar S, Al-Kabbany A, Hadhoud M (2024) An NLP-based system for modulating virtual experiences using speech instructions. Expert Syst Appl 249:123484
60. Yang Z, Liu J, Wu Z, Wu P, Liu X (2023) Video event restoration based on keyframes for video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 14592–14601
61. Li J, Li D, Savarese S, Hoi S (2023) Blip-2: bootstrapping language-image pre-training with frozen image encoders and large language models. arXiv preprint arXiv:2301.12597
62. Wu P, Liu J, He X, Peng Y, Wang P, Zhang Y (2024) Toward video anomaly retrieval from video anomaly detection: new benchmarks and model. IEEE Trans Image Process 33:2213–2225. https://doi.org/10.1109/TIP.2024.3374070
63. Khemani B, Patil S, Kotecha K, Tanwar S (2024) A review of graph neural networks: concepts, architectures, techniques, challenges, datasets, applications, and future directions. J Big Data 11(1):18
64. Cheung G, Magli E, Tanaka Y, Ng MK (2018) Graph spectral image processing. Proc IEEE 106(5):907–930

65. Shuman DI, Narang SK, Frossard P, Ortega A, Vandergheynst P (2013) The emerging field of signal processing on graphs: extending high-dimensional data analysis to networks and other irregular domains. IEEE Signal Process Mag 30(3):83–98
66. Giraldo JH, Javed S, Sultana M, Jung SK, Bouwmans T (2021) The emerging field of graph signal processing for moving object segmentation. In: International workshop on frontiers of computer vision. Springer, pp 31–45
67. Leus G, Marques AG, Moura JM, Ortega A, Shuman DI (2023) Graph signal processing: history, development, impact, and outlook. IEEE Signal Process Mag 40(4):49–60
68. Mondal A, Giraldo JH, Bouwmans T, Chowdhury AS et al (2021) Moving object detection for event-based vision using graph spectral clustering. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 876–884
69. Prummel W, Giraldo JH, Zakharova A, Bouwmans T (2023) Inductive graph neural networks for moving object segmentation. In: 2023 IEEE international conference on image processing (ICIP). IEEE, pp 2730–2734
70. Ullah W, Hussain T, Min Ullah FU, Muhammad K, Hassaballah M, Rodrigues JJ, Baik SW, Albuquerque VHCD (2023) Ad-graph: weakly supervised anomaly detection graph neural network. Int J Intell Syst 2023(1):7868415
71. Yuan H, Cai Z, Zhou H, Wang Y, Chen X (2021) Transanomaly: video anomaly detection using video vision transformer. IEEE Access 9:123977–123986
72. Nguyen T-N, Meunier J (2019) Anomaly detection in video sequence with appearance-motion correspondence. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 1273–1283
73. Majhi S, Dash R, Sa PK (2020) Temporal pooling in inflated 3dcnn for weakly-supervised video anomaly detection. In: 2020 11th International conference on computing, communication and networking technologies (ICCCNT). IEEE, pp 1–6
74. Yang Z, Liu J, Wu P (2024) Text prompt with normality guidance for weakly supervised video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 18899–18908
75. Chen J, Yang J, Wu H, Li D, Gao J, Zhou T, Xiao B (2025) Florence-vl: enhancing vision-language models with generative vision encoder and depth-breadth fusion. In: Proceedings of the computer vision and pattern recognition conference, pp 24928–24938
76. Fu D, Hua Z, Xie Y, Fang J, Zhang S, Sancak K, Wu H, Malevich A, He J, Long B (2024) Vcr-graphormer: a mini-batch graph transformer via virtual connections. In: The 12th international conference on learning representations
77. Zhou S, Shen W, Zeng D, Fang M, Wei Y, Zhang Z (2016) Spatial-temporal convolutional neural networks for anomaly detection and localization in crowded scenes. Signal Process Image Commun 47:358–368
78. Barbalau A, Ionescu RT, Georgescu M-I, Dueholm J, Ramachandra B, Nasrollahi K, Khan FS, Moeslund TB, Shah M (2023) Ssmtl++: revisiting self-supervised multi-task learning for video anomaly detection. Comput Vis Image Underst 229:103656
79. Chong YS, Tay YH (2017) Abnormal event detection in videos using spatiotemporal autoencoder. In: Advances in neural networks-ISNN 2017: 14th international symposium, ISNN 2017, Sapporo, Hakodate, and Muroran, Hokkaido, Japan, June 21–26, 2017, Proceedings, Part II. Springer, pp 189–196
80. Zeng X, Jiang Y, Ding W, Li H, Hao Y, Qiu Z (2023) A hierarchical spatio-temporal graph convolutional neural network for anomaly detection in videos. IEEE Trans Circuits Syst Video Technol 33(1):200–212. https://doi.org/10.1109/TCSVT.2021.3134410
81. Zaheer MZ, Mahmood A, Khan MH, Segu M, Yu F, Lee S-I (2022) Generative cooperative learning for unsupervised video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 14744–14754
82. Tur AO, Dall'Asen N, Beyan C, Ricci E (2023) Exploring diffusion models for unsupervised video anomaly detection. In: 2023 IEEE international conference on image processing (ICIP). IEEE, pp 2540–2544
83. Yang Z, Guo Y, Wang J, Huang D, Bao X, Wang Y (2023) Towards video anomaly detection in the real world: a binarization embedded weakly-supervised network. IEEE Trans Circuits Syst Video Technol. https://doi.org/10.1109/TCSVT.2023.3321235
84. Zaheer MZ, Lee J-h, Astrid M, Mahmood A, Lee S-I (2021) Cleaning label noise with clusters for minimally supervised anomaly detection. arXiv preprint arXiv:2104.14770
85. Zhang D, Huang C, Liu C, Xu Y (2022) Weakly supervised video anomaly detection via transformer-enabled temporal relation learning. IEEE Signal Process Lett 29:1197–1201
86. Li J, Selvaraju R, Gotmare A, Joty S, Xiong C, Hoi SCH (2021) Align before fuse: vision and language representation learning with momentum distillation. Adv Neural Inf Process Syst 34:9694–9705
87. Li D, Li J, Li H, Niebles JC, Hoi SC (2022) Align and prompt: video-and-language pre-training with entity prompts. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 4953–4963
88. Tiong AMH, Li J, Li B, Savarese S, Hoi SC (2022) Plug-and-play vqa: zero-shot vqa by conjoining large pretrained models with zero training. arXiv preprint arXiv:2210.08773
89. Antol S, Agrawal A, Lu J, Mitchell M, Batra D, Zitnick CL, Parikh D (2015) Vqa: visual question answering. In: Proceedings of the IEEE international conference on computer vision, pp 2425–2433

90. Johnson J, Hariharan B, Van Der Maaten L, Fei-Fei L, Lawrence Zitnick C, Girshick R (2017) Clevr: a diagnostic dataset for compositional language and elementary visual reasoning. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 2901–2910
91. Abdar M, Kollati M, Kuraparthi S, Pourpanah F, McDuff D, Ghavamzadeh M, Yan S, Mohamed A, Khosravi A, Cambria E et al (2024) A review of deep learning for video captioning. IEEE Trans Pattern Anal Mach Intell
92. Zeng R, Xu H, Huang W, Chen P, Tan M, Gan C (2020) Dense regression network for video grounding. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 10287–10296
93. Zhang H, Xu X, Wang X, Zuo J, Huang X, Gao C, Zhang S, Yu L, Sang N (2025) Holmes-vau: Towards long-term video anomaly understanding at any granularity. In: Proceedings of the computer vision and pattern recognition conference, pp 13843–13853
94. Ying C, Cai T, Luo S, Zheng S, Ke G, He D, Shen Y, Liu T-Y (2021) Do transformers really perform badly for graph representation? Adv Neural Inf Process Syst 34:28877–28888
95. Xu K, Hu W, Leskovec J, Jegelka S (2018) How powerful are graph neural networks? arXiv preprint arXiv:1810.00826
96. Lv H, Yue Z, Sun Q, Luo B, Cui Z, Zhang H (2023) Unbiased multiple instance learning for weakly supervised video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 8022–8031
97. Zhang C, Li G, Qi Y, Wang S, Qing L, Huang Q, Yang M-H (2023) Exploiting completeness and uncertainty of pseudo labels for weakly supervised video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 16271–16280
98. Gal Y, Ghahramani Z (2016) Dropout as a bayesian approximation: representing model uncertainty in deep learning. In: International conference on machine learning. PMLR, pp 1050–1059
99. Sabokrou M, Pourreza M, Fayyaz M, Entezari R, Fathy M, Gall J, Adeli E (2018) Avid: adversarial visual irregularity detection. In: Asian conference on computer vision. Springer, pp 488–505
100. Lin T-Y, Goyal P, Girshick R, He K, Dollár P (2017) Focal loss for dense object detection. In: Proceedings of the IEEE international conference on computer vision, pp 2980–2988
101. Tu W-C, Liu M-Y, Jampani V, Sun D, Chien S-Y, Yang M-H, Kautz J (2018) Learning superpixels with segmentation-aware affinity loss. In: Proceedings of the IEEE conference on computer vision and pattern recognition, pp 568–576
102. Kukačka J, Golkov V, Cremers D (2017) Regularization for deep learning: a taxonomy. arXiv preprint arXiv:1710.10686
103. Cho M, Kim M, Hwang S, Park C, Lee K, Lee S (2023) Look around for anomalies: weakly-supervised anomaly detection via context-motion relational learning. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 12137–12146
104. Mathieu M, Couprie C, LeCun Y (2015) Deep multi-scale video prediction beyond mean square error. arXiv preprint arXiv:1511.05440
105. Shi C, Sun C, Wu Y, Jia Y (2023) Video anomaly detection via sequentially learning multiple pretext tasks. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 10330–10340
106. Zhang M, Wang J, Qi Q, Sun H, Zhuang Z, Ren P, Ma R, Liao J (2024) Multi-scale video anomaly detection by multi-grained spatio-temporal representation learning. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 17385–17394
107. Al-lahham A, Zaheer MZ, Tastan N, Nandakumar K (2024) Collaborative learning of anomalies with privacy (clap) for unsupervised video anomaly detection: a new baseline. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR), pp 12416–12425
108. Gong D, Liu L, Le V, Saha B, Mansour MR, Venkatesh S, Hengel AVD (2019) Memorizing normality to detect anomaly: memory-augmented deep autoencoder for unsupervised anomaly detection. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 1705–1714
109. Morais R, Le V, Tran T, Saha B, Mansour M, Venkatesh S (2019) Learning regularity in skeleton trajectories for anomaly detection in videos. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 11996–12004
110. Yan C, Zhang S, Liu Y, Pang G, Wang W (2023) Feature prediction diffusion model for video anomaly detection. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 5527–5537
111. Ristea N-C, Croitoru F-A, Ionescu RT, Popescu M, Khan FS, Shah M et al (2024) Self-distilled masked auto-encoders are efficient video anomaly detectors. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 15984–15995
112. Cao C, Zhang H, Lu Y, Wang P, Zhang Y (2024) Scene-dependent prediction in latent space for video anomaly detection and anticipation. IEEE Trans Pattern Anal Mach Intell
113. Fioresi J, Dave IR, Shah M (2023) Ted-spad: temporal distinctiveness for self-supervised privacy-preservation for video anomaly detection. In: Proceedings of the IEEE/CVF international conference on computer vision, pp 13598–13609

114. Chen J, Li L, Su L, Zha Z-j, Huang Q (2024) Prompt-enhanced multiple instance learning for weakly supervised video anomaly detection. In: Proceedings of the IEEE/CVF conference on computer vision and pattern recognition, pp 18319–18329
115. Song I, Lee S, Joo M, Lee J (2025) Anomaly detection for people with visual impairments using an egocentric 360-degree camera. In: 2025 IEEE/CVF winter conference on applications of computer vision (WACV). IEEE, pp 2828–2837
116. Hinami R, Mei T, Satoh S (2017) Joint detection and recounting of abnormal events by learning deep generic knowledge. In: Proceedings of the IEEE international conference on computer vision, pp 3619–3627
117. Wu C, Shao S, Tunc C, Satam P, Hariri S (2022) An explainable and efficient deep learning framework for video anomaly detection. Cluster Comput 1–23
118. Lundberg S (2017) A unified approach to interpreting model predictions. arXiv preprint arXiv:1705.07874
119. Eck N, Waltman L (2010) Software survey: vosviewer, a computer program for bibliometric mapping. Scientometrics 84(2):523–538
120. Awasthi A, Ly S, Nizam J, Zare S, Mehta V, Ahmed S, Shah K, Nemani R, Prasad S, Van Nguyen H (2023) Anomaly detection in satellite videos using diffusion models. arXiv preprint arXiv:2306.05376
121. Batzner K, Heckler L, König R (2024) Efficientad: accurate visual anomaly detection at millisecond-level latencies. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision, pp 128–138
122. Nie Y, Huang H, Long C, Zhang Q, Maji P, Cai H (2024) Interleaving one-class and weakly-supervised models with adaptive thresholding for unsupervised video anomaly detection. arXiv preprint arXiv:2401.13551
123. Zanella L, Liberatori B, Menapace W, Poiesi F, Wang Y, Ricci E (2023) Delving into clip latent space for video anomaly recognition. arXiv preprint arXiv:2310.02835
124. Sethi A, Saini K, Mididoddi SM (2023) Video anomaly detection using GAN
125. Aafaq N, Mian A, Liu W, Gilani SZ, Shah M (2019) Video description: a survey of methods, datasets, and evaluation metrics. ACM Comput Surv (CSUR) 52(6):1–37

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